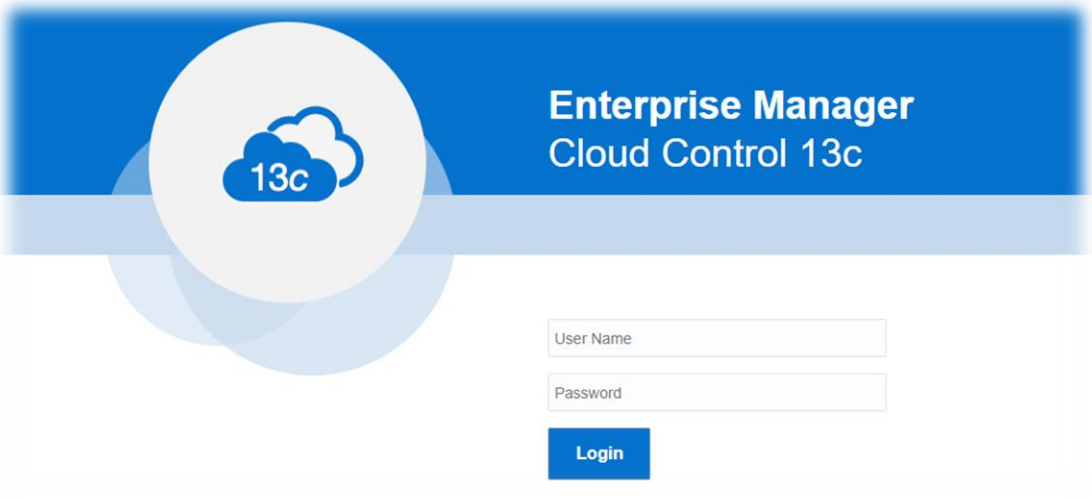


Oracle Enterprise Manager (OEM) 13c Installation & Configuration

Complete Guide for Oracle RAC Monitoring

(step-by step Implementation)

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1. Introduction

Oracle Enterprise Manager Cloud Control is Oracle's centralized management platform for monitoring, administering, and managing Oracle databases and the underlying IT infrastructure from a single web-based console. It provides DBAs with a unified view of database availability, performance, configuration, host resources, and other managed targets across an enterprise environment.

Oracle Enterprise Manager Cloud Control 13c uses a multi-tier architecture consisting of the **Enterprise Manager Cloud Control console, Oracle Management Service (OMS), Management Agents, and a Management Repository**. The Management Repository stores monitoring and management information collected from the targets, while the OMS processes this information and provides the central management interface through the Enterprise Manager console.

In an Oracle database environment, Enterprise Manager can be used to centrally monitor individual databases as well as clustered environments such as **Oracle Real Application Clusters (RAC)**. Management Agents installed on target hosts collect database and operating-system information and communicate with the OMS. This allows DBAs to monitor target availability, performance metrics, alerts, resource utilization, and configuration from a centralized platform.

The **Management Repository** is a critical component of the Enterprise Manager architecture. It stores information related to discovered targets, monitoring data, configuration information, alerts, jobs, and other management activities. In this implementation, an Oracle Database 19c database named **emrep** is used as the Enterprise Manager Management Repository.

This document provides a complete, hands-on, step-by-step guide for installing and configuring **Oracle Enterprise Manager Cloud Control 13c Release 5 (13.5.0.0.0)** in an **Oracle Linux environment running on VMware Workstation**. The implementation covers Management Repository preparation, Enterprise Manager software installation, OMS configuration, Management Agent setup, connectivity validation, and Oracle RAC target discovery.

The implementation also demonstrates the integration of Enterprise Manager with a **two-node Oracle RAC environment**, where Management Agents are configured on the RAC hosts and the RAC environment is discovered and registered through the Enterprise Manager console.

Enterprise Manager Architecture

The major components used in this implementation are:

- **Enterprise Manager Cloud Control Console** : Web-based interface used by DBAs to monitor and administer targets.

- **Oracle Management Service (OMS)** : Central application tier that communicates between the Enterprise Manager console, Management Agents, and Management Repository.
- **Management Agent** : Installed on managed hosts to collect monitoring and configuration information and communicate with OMS.
- **Management Repository** : Oracle Database repository that stores Enterprise Manager management data.
- **Oracle RAC Targets** : Two-node RAC environment monitored and managed through Enterprise Manager.

How Management Agents Work

Management Agents are installed on the hosts that need to be monitored. The agents collect information from databases, hosts, and other configured targets and upload the collected data to the OMS.

The general communication flow is:

Managed Target → Management Agent → Oracle Management Service → Management Repository

The Enterprise Manager console communicates with the OMS, allowing administrators to view the collected information and perform supported management operations.

Agent health is therefore an important part of the Enterprise Manager architecture. Agent status, upload status, heartbeat, collection status, and connectivity must be validated to ensure that monitoring information is being successfully transferred to the OMS.

Oracle RAC Monitoring

Oracle Enterprise Manager provides centralized visibility into Oracle RAC environments. After the RAC hosts and database targets are discovered, DBAs can monitor the cluster, instances, databases, hosts, and related components through the Enterprise Manager console.

In this implementation, Management Agents are configured on **node1** and **node2**, and the RAC environment is subsequently discovered through Enterprise Manager. The implementation validates agent communication, uploads, heartbeat, collections, and target discovery.

Why Use Oracle Enterprise Manager Cloud Control

Enterprise Manager provides several advantages for centralized Oracle environment management:

- **Centralized monitoring** of Oracle databases and infrastructure.
- **RAC monitoring** from a single management console.
- **Proactive alerting** for database and host conditions.
- **Performance monitoring** using collected database and infrastructure metrics.
- **Centralized target discovery and management**.
- **Management Agent-based monitoring** across multiple hosts.
- **Historical monitoring data** stored in the Management Repository.
- **Improved operational visibility** for DBA and infrastructure teams.
- **Reduced manual effort** compared with monitoring individual databases independently.

What This Guide Covers

This document presents a complete, practical implementation of **Oracle Enterprise Manager Cloud Control 13c Release 5 (13.5.0.0.0)** with an **Oracle Database 19c Management Repository** and monitoring of a **two-node Oracle RAC environment**.

It covers the implementation from the initial environment and prerequisite validation through Management Repository preparation, Enterprise Manager installation, OMS configuration, root-level configuration, Management Agent setup, SSH connectivity, agent validation, and RAC target discovery.

The guide combines **command-line implementation with GUI-based Enterprise Manager configuration and validation**, providing both the technical commands and corresponding operational workflow. It is intended as a practical reference for **Oracle DBAs, database administrators, infrastructure administrators, and learners** who want hands-on experience installing Enterprise Manager Cloud Control and using it to monitor an Oracle RAC environment.

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2. Objective / Scope

Objective

- Install Oracle Database 19c software required for the Enterprise Manager Management Repository.
- Create and validate the emrep repository database.
- Install Oracle Enterprise Manager Cloud Control 13c Release 5.
- Configure the OMS, repository connection, management repository tablespace, and required services.

- Establish secure SSH key-based connectivity from the OEM host to managed RAC hosts.
- Deploy and validate Oracle Management Agent 13.5.0.0.0 on node1 and node2.
- Discover and register the RAC environment through the OEM console.
- Validate agent heartbeat, upload, collection status, and target readiness.
- Document the implementation with both GUI screenshots and actual command execution evidence.

Scope

- OEM host preparation and required operating-system configuration.
- Oracle Database 19c repository installation and configuration.
- Enterprise Manager 13c Release 5 installation and initial configuration.
- Management Agent deployment and health validation.
- Oracle RAC target discovery and registration.
- Post-installation verification and operational observations.

Out of Scope

- Enterprise Manager high availability using multiple OMS servers.
- Advanced BI Publisher configuration and reporting customization.
- Enterprise Manager Cloud Control upgrades between major releases.
- Production sizing recommendations based on this lab environment.
- Licensed Management Pack features not explicitly demonstrated in this implementation.

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3. Architecture Overview

Layer	Component	Role
Management	OEM.localdomain	Hosts OMS, repository connection and Enterprise Manager services
Repository	Oracle 19c emrep	Stores Enterprise Manager management data
Agent	node1	Managed host with OEM 13.5 Management Agent
Agent	node2	Managed host with OEM 13.5 Management Agent
Database	Oracle RAC	Target database / cluster monitored by OEM

Administration	Enterprise Manager Console	Central GUI for target discovery and monitoring

Logical Flow: Administrator → Enterprise Manager Console → OMS → Management Agents → RAC hosts / database targets.

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4. Environment Details

Parameter	Value
OEM Hostname	OEM.localdomain
OEM Host IP / Network	Lab environment; hostname-based connectivity used
Operating System	Linux x86-64 (UEK kernel observed on managed hosts)
Repository Database	Oracle Database 19c Enterprise Edition
Repository SID / Global DB	emrep
Repository Oracle Home	/u01/app/oracle/product/19.0.0/dbhome_1
Oracle Base	/u01/app/oracle
Repository Storage	Filesystem
Repository Datafiles	/u02/oradata
Management Repository Tablespace	mgmt_tablespace
Enterprise Manager Release	13.5.0.0.0 (Cloud Control 13c Release 5)
OMS Host	OEM.localdomain
Middleware Home	/u01/app/oracle/middleware
Agent Base	/u01/app/oracle/agent
Agent Home	/u01/app/oracle/product/agent13c/agent_inst
Managed Hosts	node1.localdomain / node2.localdomain
Agent Port	3872
Repository / OMS Ports Observed	4903 / 4904 / 7802
Oracle Listener Port	1521

Enterprise Manager Console	https://OEM.localdomain:7803/em
Admin Server	https://OEM.localdomain:7102/console
RAC Targets	Oracle RAC cluster, host and Clusterware-related targets

NOTE :- The exact operating-system and storage sizing values in this table reflect the supplied implementation and should not be treated as production sizing guidance.

---

5. Prerequisites

Before starting the installation, ensure the following prerequisites are met:

• Root/sudo access on the OEM node, and oracle-user SSH access to both RAC nodes

• All 5 Enterprise Manager 13.5 installer files downloaded and checksum-verified

• Oracle Database 19c software (LINUX.X64_193000_db_home.zip) available for the repository database

• Sufficient disk space on /u01 and /u02 (verified: 112 GB each on OEM node)

• Name resolution for OEM, node1, node2, VIPs, and SCAN entries present in /etc/hosts on all 3 hosts

• X11 forwarding available on the OEM node (DISPLAY=:0) if using the graphical installer

• Firewall and SELinux posture on the OEM node reviewed and adjusted (OEM node runs Enforcing/active by default, unlike the RAC nodes)

• Required downtime / change approval obtained, if applicable

---

6. Implementation Steps

The Oracle Enterprise Manager 13c implementation was performed in multiple phases. The OEM server was first prepared as the Management Server and Repository host. Oracle Database 19c was installed as the Management Repository database, followed by Enterprise Manager 13.5 installation. After OMS configuration, Management Agents were deployed to both RAC nodes, and the existing Oracle RAC environment was discovered and added to Enterprise Manager.

The implementation was divided into the following phases:

PHASE 1 : OS and User Preparation (OEM Node)

Objective : The objective of this phase is to prepare the OEM server with the required Oracle software prerequisites, directory structure, security configuration, and environment variables.

The OEM server acts as the central Enterprise Manager server and hosts the Management Repository database and Oracle Management Service (OMS).

Update the Operating System, Install Oracle Database Prerequisites & Configure the Oracle User

```
[root@OEM ~]# yum update -y
[root@OEM ~]# yum install -y oracle-database-preinstall-19c
[root@OEM ~]# passwd oracle
Changing password for user oracle.
New password:
BAD PASSWORD: The password is shorter than 8 characters

Retype new password:
passwd: all authentication tokens updated successfully.
[root@OEM ~]#
```

This ensures that the server has the latest available operating-system packages before beginning the Oracle software installation. The preinstallation package prepares the operating system with many of the required users, groups, kernel parameters, limits, and prerequisite packages required for Oracle Database. The oracle account is used to install and operate the Oracle Database software and Enterprise Manager components.

Create the Required Directory Structure

Create the Oracle software, inventory, middleware, agent, software-library, and database storage directories and Set the correct ownership and permissions.

```
[root@OEM ~]# mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1
[root@OEM ~]# mkdir -p /u01/app/oraInventory
[root@OEM ~]# mkdir -p /u01/app/oracle/middleware
[root@OEM ~]# mkdir -p /u01/app/oracle/agent
[root@OEM ~]# mkdir -p /u01/app/oracle/swlib
[root@OEM ~]# mkdir -p /u02/oradata
[root@OEM ~]# chown -R oracle:oinstall /u01 /u02
[root@OEM ~]# chmod -R 775 /u01 /u02
[root@OEM ~]#
```

Configure SELinux

For this lab implementation, SELinux was changed to permissive mode & Verify the configuration:

```
[root@OEM ~]# setenforce 0
[root@OEM ~]# sed -i 's/^SELINUX=enforcing/SELINUX=permissive/' /etc/selinux/config
[root@OEM ~]# cat /etc/selinux/config

# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#     enforcing - SELinux security policy is enforced.
```

```
#     permissive - SELinux prints warnings instead of enforcing.
#     disabled - No SELinux policy is loaded.
SELINUX=permissive
# SELINUXTYPE= can take one of three values:
#     targeted - Targeted processes are protected,
#     minimum - Modification of targeted policy. Only selected processes are protected.
#     mls - Multi Level Security protection.
SELINUXTYPE=targeted

[root@OEM ~]#
```

For production environments, SELinux requirements should be reviewed according to the organization's security standards and Oracle's supported configuration requirements.

Configure Firewall Ports

Open the ports required by the repository database, OMS, and Management Agent.

```
[root@OEM ~]# firewall-cmd --permanent --add-port=1521/tcp
success
[root@OEM ~]# firewall-cmd --permanent --add-port=4903/tcp
success
[root@OEM ~]# firewall-cmd --permanent --add-port=4904/tcp
success
[root@OEM ~]# firewall-cmd --permanent --add-port=7802/tcp
success
[root@OEM ~]# firewall-cmd --permanent --add-port=3872/tcp
success
[root@OEM ~]# firewall-cmd --reload
success
[root@OEM ~]#
```

These ports are used by the implementation for Oracle Net connectivity, OMS communication, console-related services, and Management Agent communication.

Configure Oracle Environment Variables

Switch to the Oracle user & Create an environment file:

```
[root@OEM ~]# su - oracle
[oracle@OEM ~]$
[oracle@OEM ~]$ vi db_env
[oracle@OEM ~]$ cat db_env
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1
export PATH=$ORACLE_HOME/bin:$PATH
[oracle@OEM ~]$
```

The same environment configuration was used during the actual implementation.

PHASE 2 : Install Oracle 19c Database Software (Repository Home)

Objective : Enterprise Manager requires a Management Repository database. In this implementation, Oracle Database 19c was installed on the OEM server and later used to create the emrep repository database.

Stage the Oracle Database Software

The Oracle Database 19c software ZIP was placed in:

```
[oracle@OEM ~]$ cd /u01/app/oracle/swlib/
[oracle@OEM swlib]$ ls
LINUX.X64_193000_db_home.zip
[oracle@OEM swlib]$
```

Extract the Oracle Home

This populates the Oracle Home with the Database 19c binaries.

```
[oracle@OEM ~]$ cd $ORACLE_HOME
[oracle@OEM dbhome_1]$
[oracle@OEM dbhome_1]$ unzip /u01/app/oracle/swlib/LINUX.X64_193000_db_home.zip
```

Start the Oracle Universal Installer

Run:

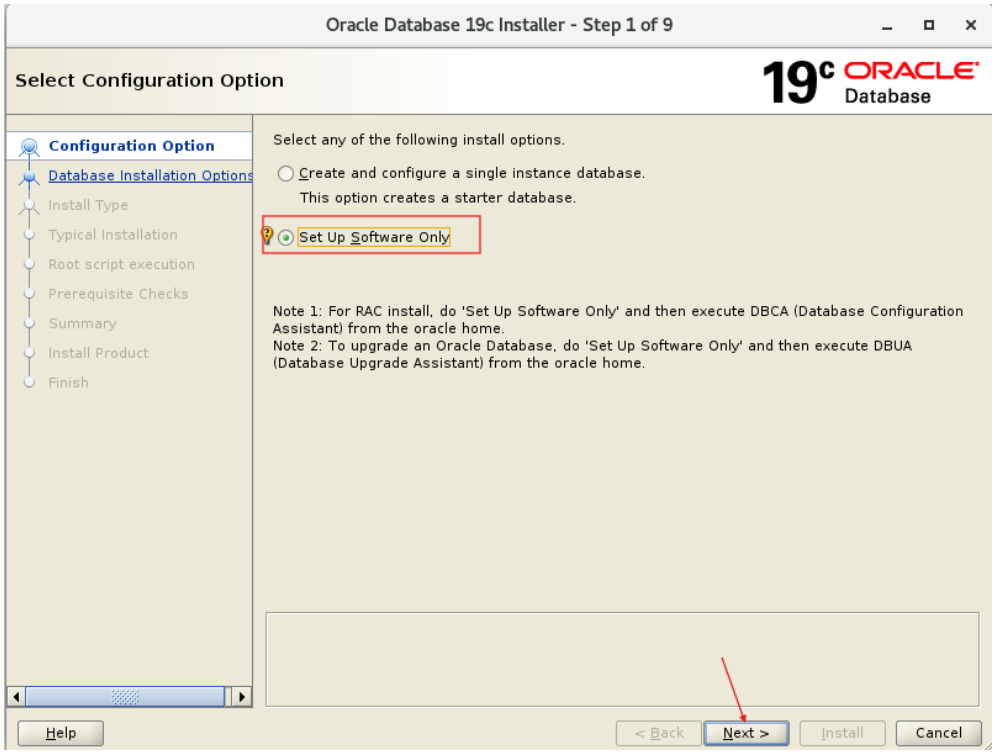
```
[oracle@OEM dbhome_1]$ ./runInstaller
Launching Oracle Database Setup Wizard...
```

The Oracle Universal Installer starts the Database software installation wizard.

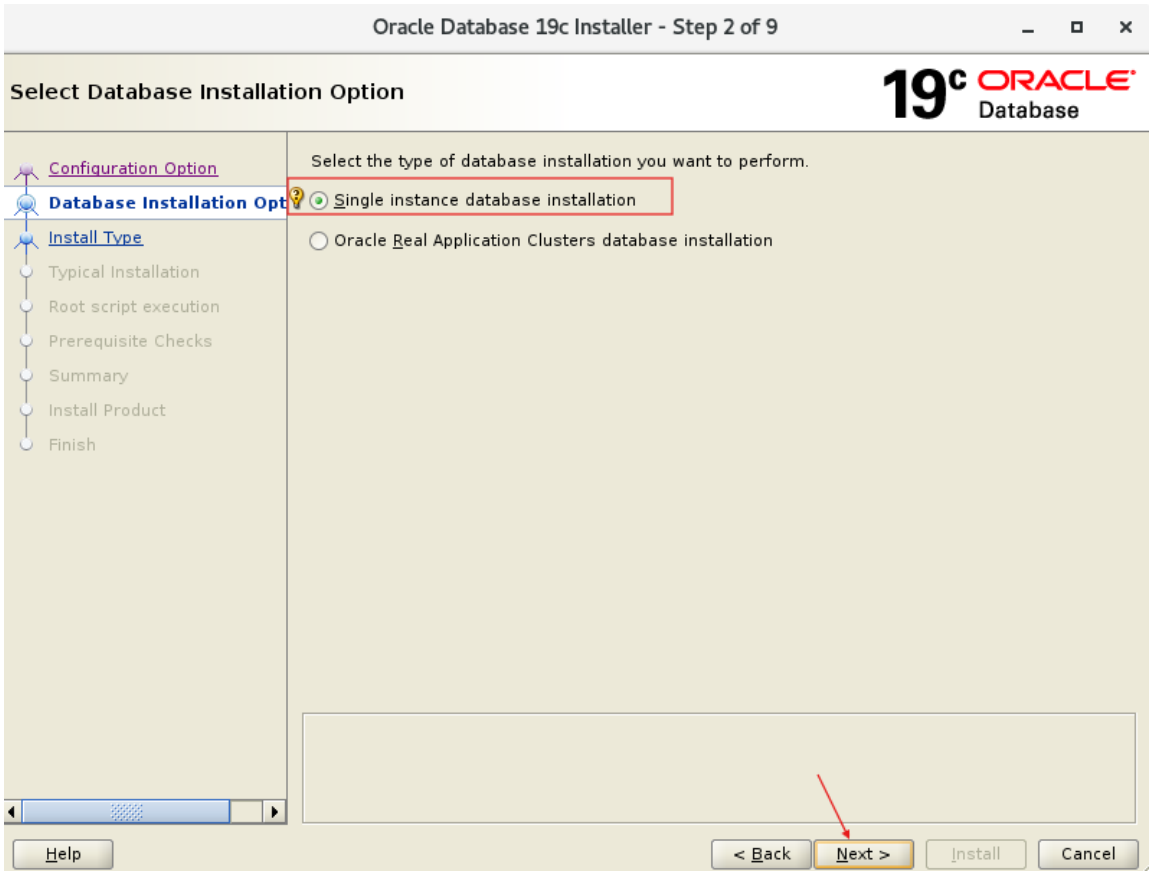
For this implementation, install the Oracle Database software only. The database itself will be created separately using DBCA.

The actual installation session was completed successfully and generated the Oracle installation response and log files.

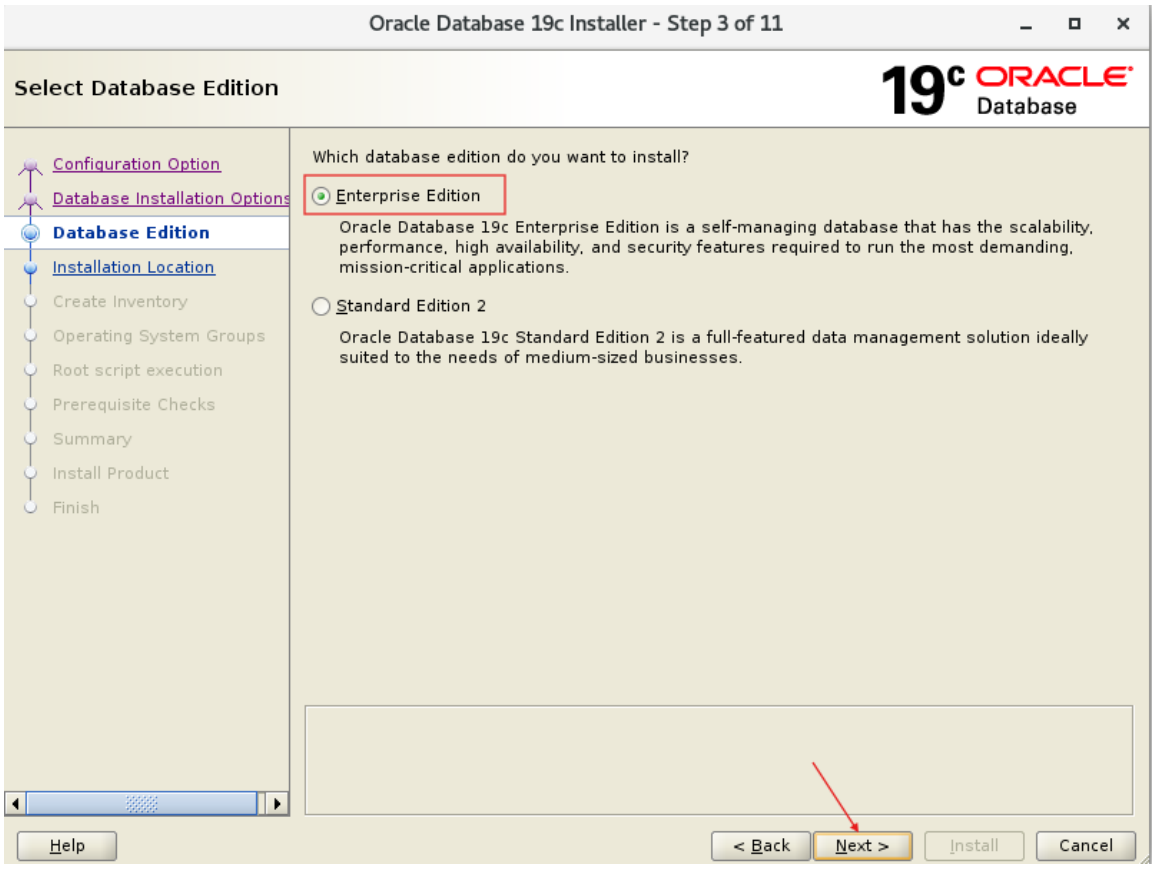
Choose **Set Up Software Only** (not "Create and configure a database". you'll create the repository DB separately with dbca). Click **Next**.



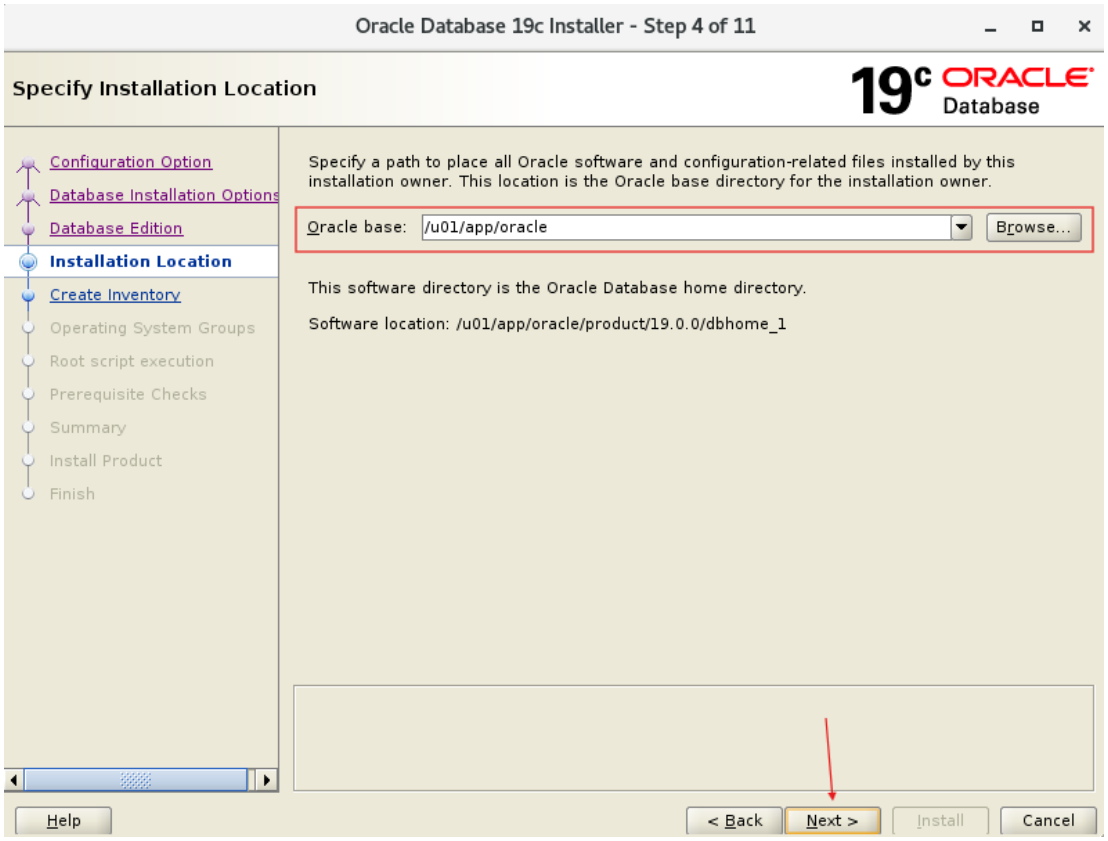
Choose **Single instance database installation**. Click **Next**.



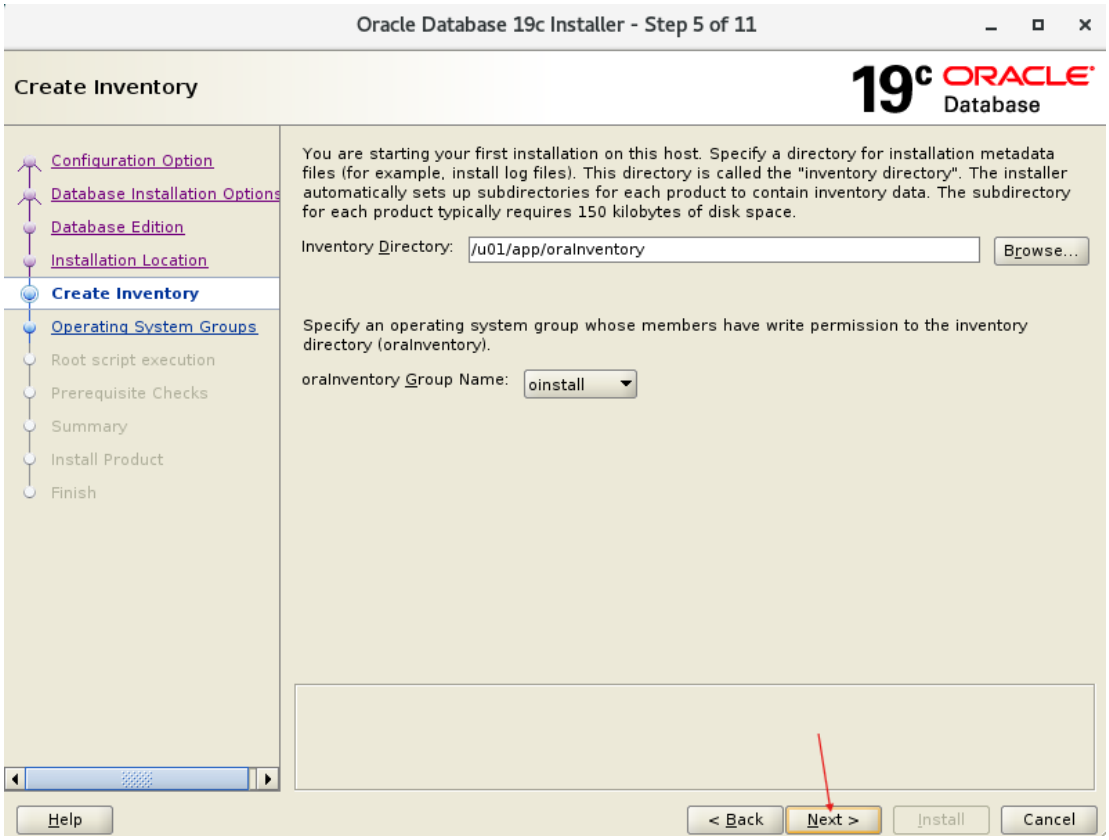
Choose **Enterprise Edition**. Click **Next**.



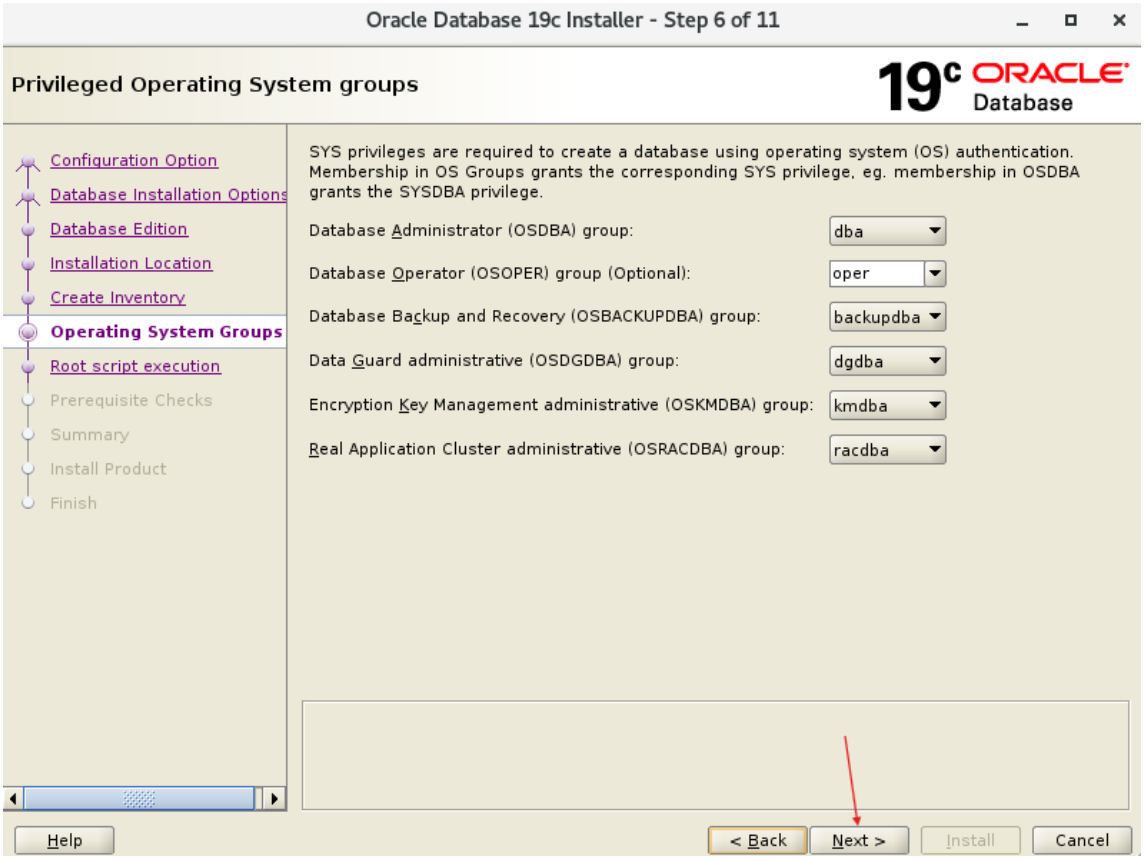
Oracle base: /u01/app/oracle. Software location auto-fills as /u01/app/oracle/product/19.0.0/dbhome_1. Click **Next**.



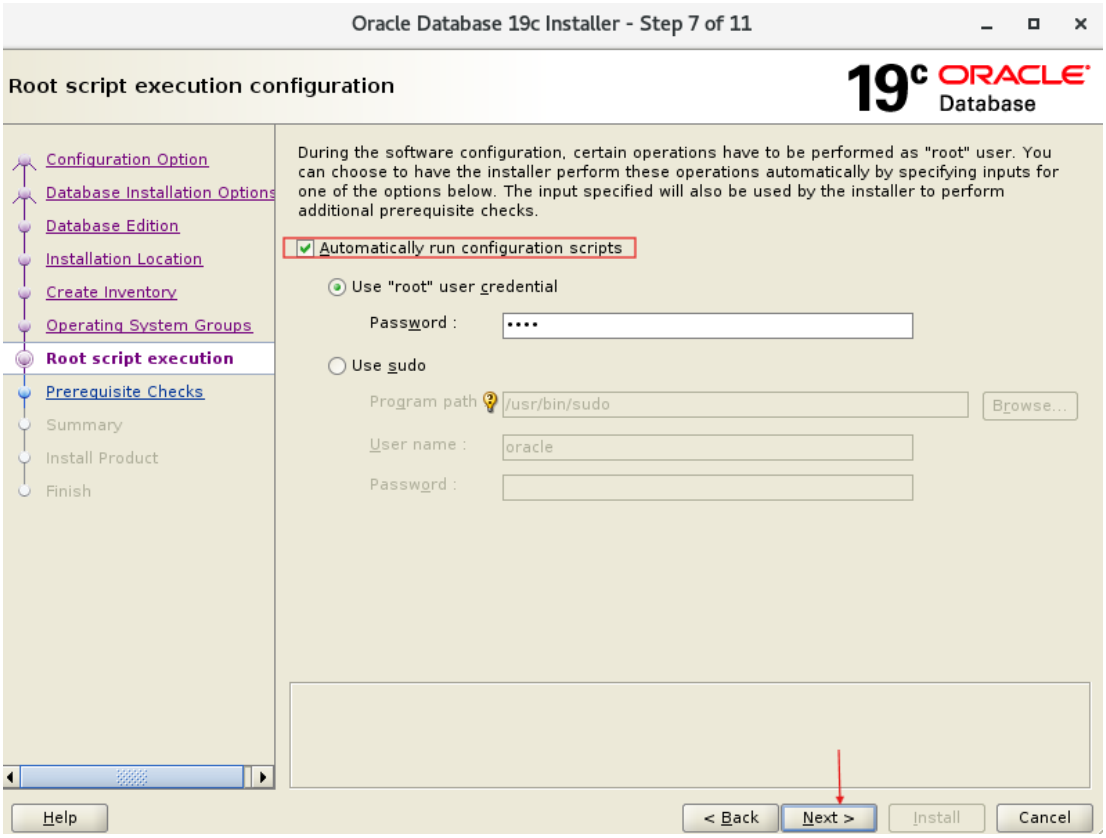
Inventory Directory: /u01/app/oraInventory, Group: oinstall. Click **Next**.



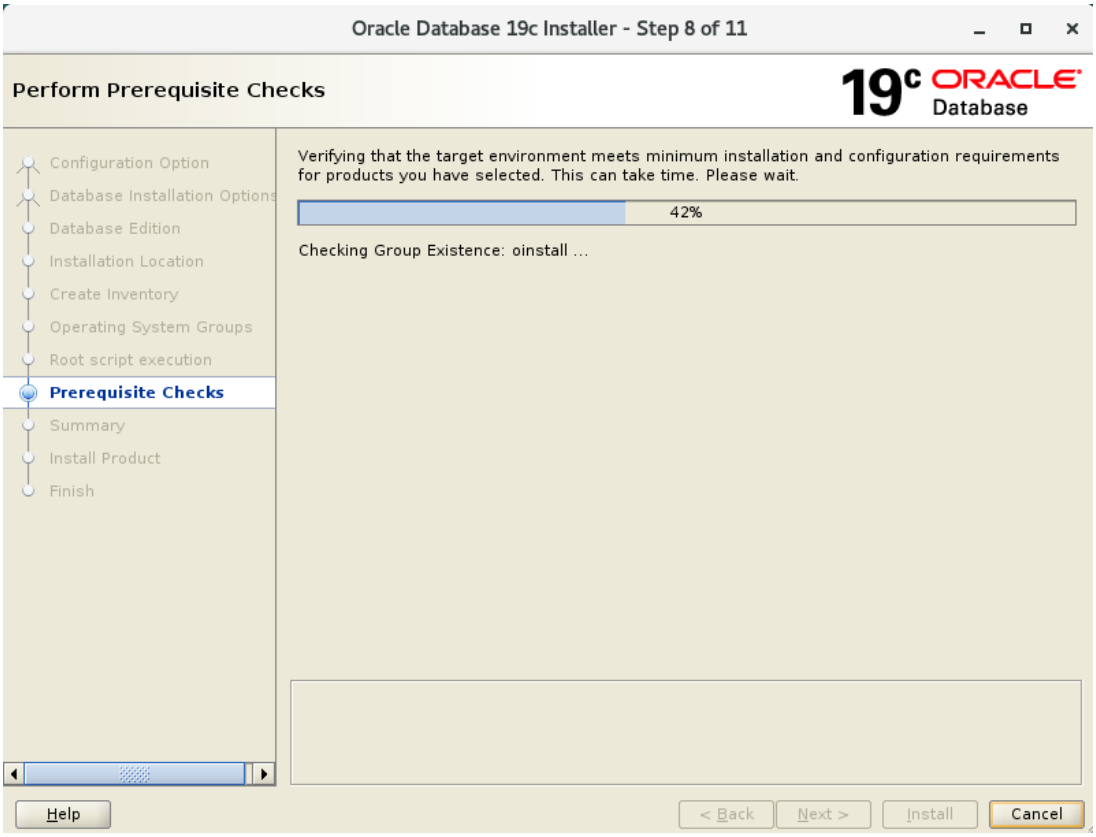
Confirm OSDBA=dba, OSOPER=oper, OSBACKUPDBA=backupdba, OSDGDBA=dgdba, OSKMDBA=kmdba, OSRACDBA=racdba. Click **Next**.



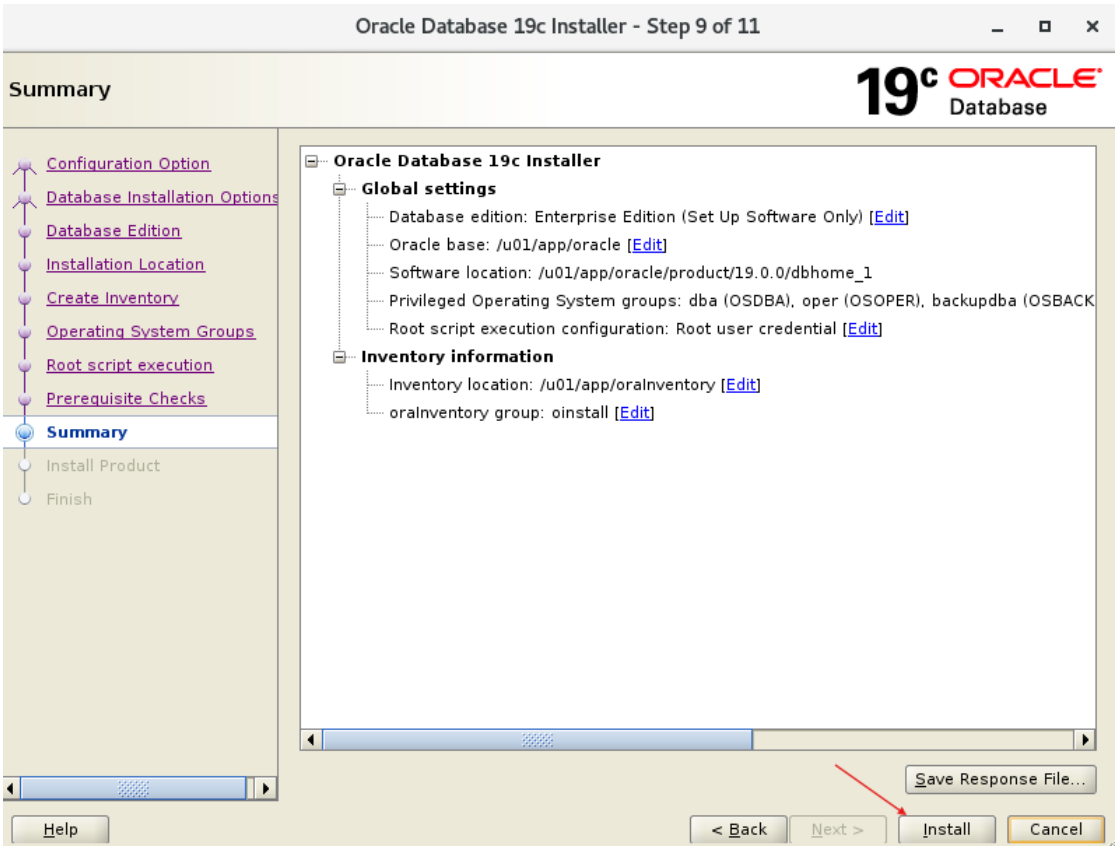
Check **Automatically run configuration scripts** as for testing environment, select **Use "root" user credential**, enter the root password. Click **Next**



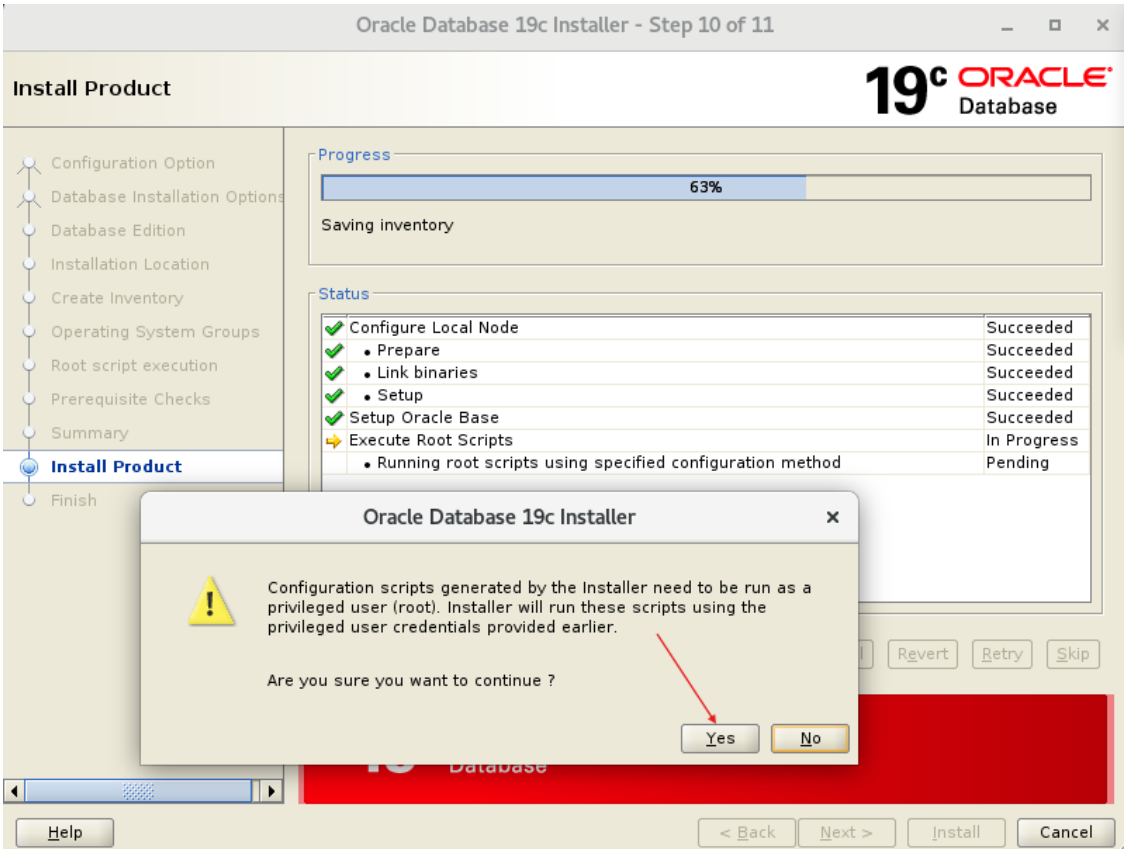
Installer auto-verifies OS packages, kernel params, memory, swap. Wait for it to complete, resolve any failures shown. Click **Next**.



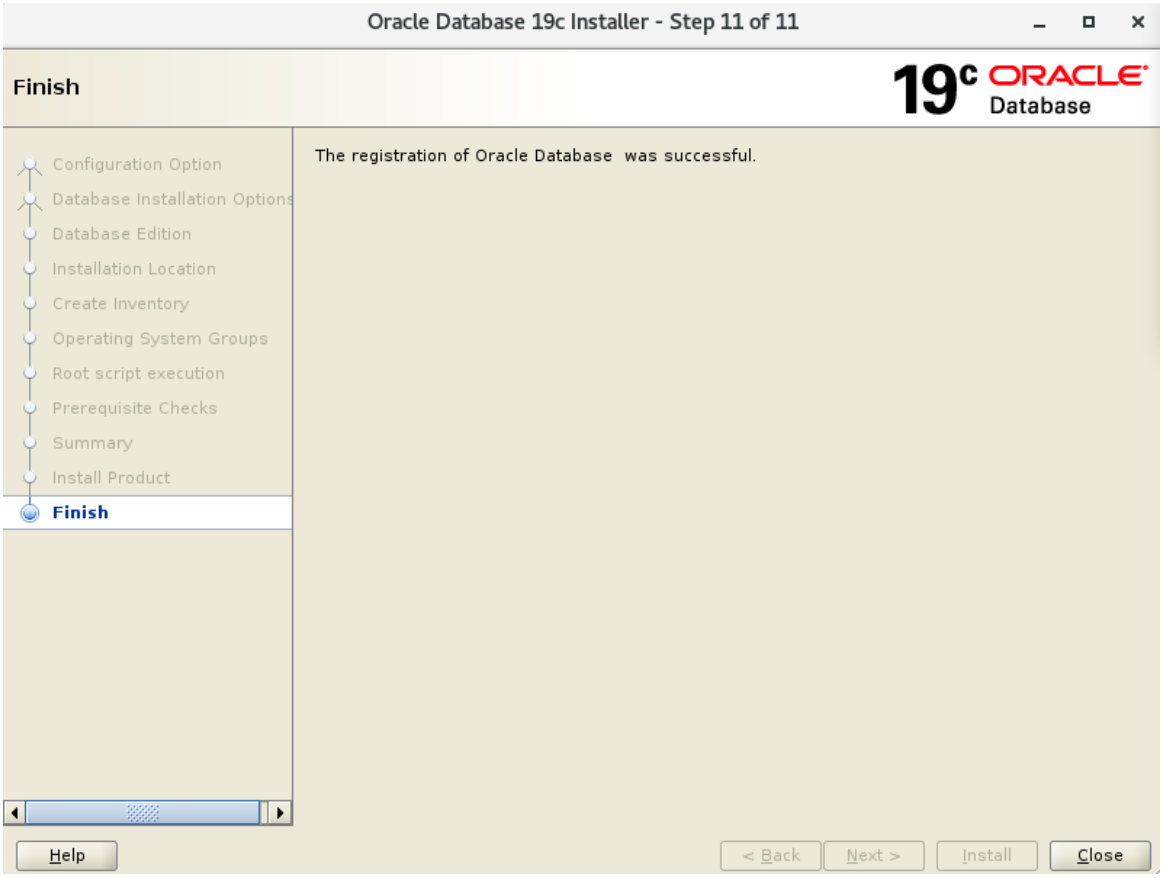
Review the settings (edition, base, root script config). Click **Install**.



Progress bar runs; a popup asks to confirm running root scripts automatically. click **Yes**.



Confirms "The registration of Oracle Database was successful." Click **Close**.



PHASE 3 : Create the OEM Management Repository Database

Objective : The Management Repository is the database used by Enterprise Manager to store monitoring data, target information, configuration information, jobs, events, and other Enterprise Manager metadata.

In this implementation, the repository database was named:

Database Name : emrep

SID : emrep

Create the Repository Database

Use DBCA to create the repository database.

```
[oracle@OEM dbhome_1]$ dbca -silent -createDatabase \
> -templateName General_Purpose.dbc \
> -gdbname emrep \
> -sid emrep \
> -characterSet AL32UTF8 \
> -sysPassword 'Welcome$123' \
> -systemPassword 'Welcome$123' \
> -createAsContainerDatabase false \
> -databaseType MULTIPURPOSE \
> -totalMemory 2048 \
> -storageType FS \
> -datafileDestination /u02/oradata \
> -listeners LISTENER \
> -emConfiguration NONE \
> -variables "ORACLE_BASE=/u01/app/oracle" \
> -ignorePreReqs
Prepare for db operation
10% complete
Copying database files
40% complete
Creating and starting Oracle instance
42% complete
46% complete
50% complete
54% complete
60% complete
Completing Database Creation
66% complete
69% complete
70% complete
Executing Post Configuration Actions
100% complete
Database creation complete. For details check the logfiles at:
 /u01/app/oracle/cfgtoollogs/dbca/emrep.
Database Information:
Global Database Name:emrep
System Identifier(SID):emrep
Look at the log file "/u01/app/oracle/cfgtoollogs/dbca/emrep/emrep.log" for further details.
[oracle@OEM dbhome_1]$
```

Replace the password placeholders with the passwords selected for the environment.

The implementation created a non-CDB database named emrep, using filesystem storage under /u02/oradata.

Verify the Repository Database

```
[oracle@OEM dbhome_1]$ ps -ef|grep pmon
oracle   15920     1  0 13:02 ?        00:00:00 ora_pmon_emrep
oracle   16320  3025  0 13:03 pts/0    00:00:00 grep --color=auto pmon
[oracle@OEM dbhome_1]$
[oracle@OEM dbhome_1]$ lsnrctl status

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 14-SEP-2026 13:03:38

Copyright (c) 1991, 2019, Oracle.  All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=OEM.localdomain)(PORT=1521)))
STATUS of the LISTENER
------------------------
Alias                     LISTENER
Version                   TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date                14-SEP-2026 12:46:18
Uptime                    0 days 0 hr. 17 min. 19 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora
Listener Log File         /u01/app/oracle/diag/tnslsnr/OEM/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=OEM.localdomain)(PORT=1521)))
Services Summary...
Service "emrep" has 1 instance(s).
  Instance "emrep", status READY, has 1 handler(s) for this service...
Service "emrepXDB" has 1 instance(s).
  Instance "emrep", status READY, has 1 handler(s) for this service...
The command completed successfully
[oracle@OEM dbhome_1]$
```

The implementation verified both the emrep instance and its listener service.

Verify Database Connectivity

```
[oracle@OEM dbhome_1]$ export ORACLE_SID=emrep
[oracle@OEM dbhome_1]$
[oracle@OEM dbhome_1]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Mon Sep 14 13:05:42 2026
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle.  All rights reserved.


Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> set line 200 pages 200
col host_name for a30
col Instance_name for a30
select distinct instance_name,Host_name,open_mode,database_role,log_mode,startup_time,created,edition
from gv$instance,gv$database;

INSTANCE_NAME                HOST_NAME                      OPEN_MODE            DATABASE_ROLE    LOG_MODE     STARTUP_T CREATED   EDITION
---------------------------- ------------------------------ -------------------- ---------------- ------------ --------- --------- -------
emrep                        OEM.localdomain                READ WRITE           PRIMARY          NOARCHIVELOG 14-SEP-26 14-SEP-26 EE
```

This confirms that the repository database is available and operating correctly before Enterprise Manager installation.

Create the Management Tablespace

Create a dedicated tablespace for Enterprise Manager management data.

```
SQL> CREATE TABLESPACE mgmt_tablespace
  DATAFILE '/u02/oradata/EMREP/mgmt_tablespace01.dbf' SIZE 10G AUTOEXTEND ON NEXT 500M MAXSIZE 30G;

Tablespace created.

SQL>
```

This provides dedicated storage for management-related repository data.

Configure Repository Database Parameters

Configure the required database parameters:

```
SQL> ALTER SYSTEM SET processes=600 SCOPE=SPFILE;

System altered.

SQL> ALTER SYSTEM SET open_cursors=1000 SCOPE=SPFILE;

System altered.

SQL> ALTER SYSTEM SET session_cached_cursors=200 SCOPE=SPFILE;

System altered.

SQL> ALTER SYSTEM SET "_allow_insert_with_update_check"=TRUE SCOPE=SPFILE;

System altered.

SQL>
```

These parameters are stored in the server parameter file and therefore require a database restart.

```
SQL> SHUTDOWN IMMEDIATE;
Database closed.
Database dismounted.
ORACLE instance shut down.
SQL>
SQL> STARTUP;
ORACLE instance started.

Total System Global Area 1610609200 bytes
Fixed Size                  8897072 bytes
Variable Size             385875968 bytes
Database Buffers         1207959552 bytes
Redo Buffers                7876608 bytes
Database mounted.
Database opened.

SQL>
```

PHASE 4 : Stage Enterprise Manager 13.5 Software

Objective

This phase prepares the Enterprise Manager 13.5 installation media before starting the OMS installation.

Verify Enterprise Manager Installation Files

```
[oracle@OEM dbhome_1]$ cd /u01/app/oracle/swlib/
[oracle@OEM swlib]$
[oracle@OEM swlib]$ ls -ltrh
total 11G
-rw-r--r--. 1 oracle oinstall 2.9G Jun 16  2025 LINUX.X64_193000_db_home.zip
-rw-r--r--. 1 oracle oinstall 1.5G Sep 13 17:11 em13500_linux64.bin
-rw-r--r--. 1 oracle oinstall 1.4G Sep 13 17:15 em13500_linux64-5.zip
-rw-r--r--. 1 oracle oinstall 2.0G Sep 13 17:17 em13500_linux64-3.zip
-rw-r--r--. 1 oracle oinstall 1.8G Sep 13 17:18 em13500_linux64-2.zip
-rw-r--r--. 1 oracle oinstall 1.4G Sep 13 17:18 em13500_linux64-4.zip
[oracle@OEM swlib]$
```

All installation files were staged in the same software-library directory.

Verify File Integrity

Calculate SHA-256 checksums:

```
[oracle@OEM swlib]$ sha256sum em13500_linux64*
d6ba033bd0684b9c5b2b544c02e1b65378e5ac4b11e7373022eac79664500221  em13500_linux64-2.zip
52c368056df792b6e605c83b42cb94460fce729f2201d4438ea3d62df25bbf10  em13500_linux64-3.zip
c5fd4becc5cd1ef51c43855250eb89924995c4cd311f588770ffa14a2d543ed0  em13500_linux64-4.zip
f7f2fc7a8cd71f48023f11efc24857e98cc460e2aebe5fe231ec050446d34c65  em13500_linux64-5.zip
dcae63f7759ce00ad90b20dd32756015a783dc98ef1e5b039bda5775634b7bbc  em13500_linux64.bin
[oracle@OEM swlib]$
```

This provides a basic integrity check of the downloaded Enterprise Manager installation media before installation.

Make the Installer Executable

```
[oracle@OEM swlib]$ chmod +x em13500_linux64.bin
[oracle@OEM swlib]$
```

The installer should have executable permission before launching it.

PHASE 5 : Install Oracle Enterprise Manager 13.5 OMS

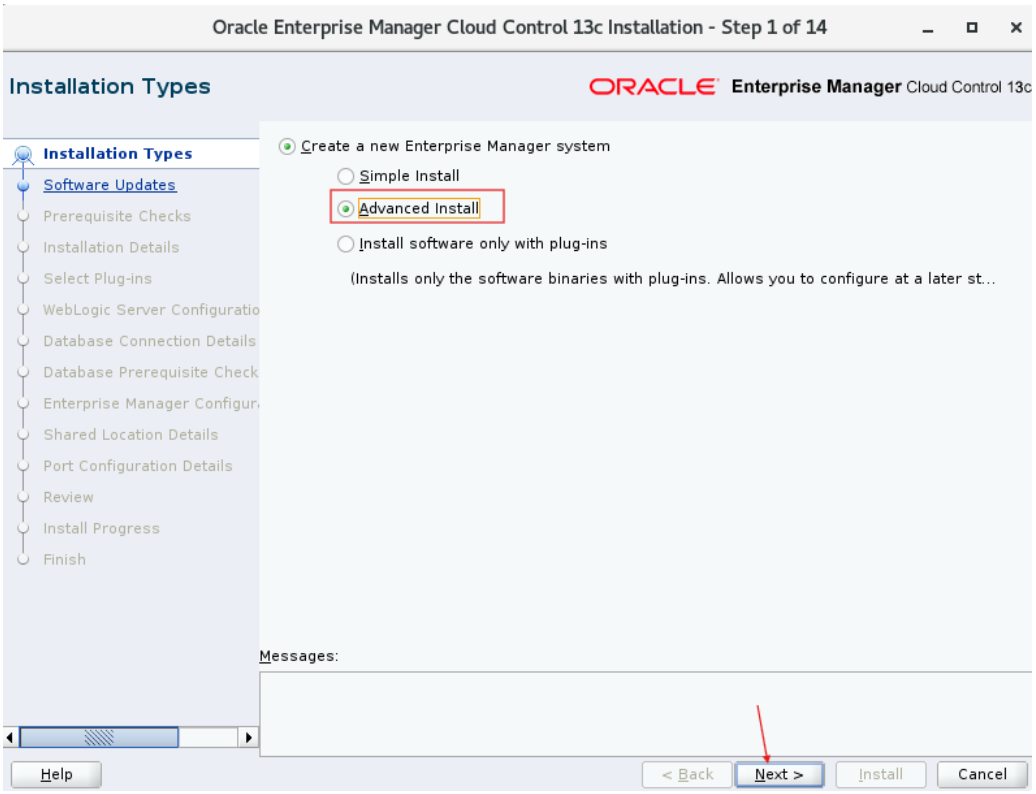
Objective : This phase installs the Oracle Management Service and configures the Enterprise Manager system using the previously created emrep repository database.

Launch the Enterprise Manager Installer

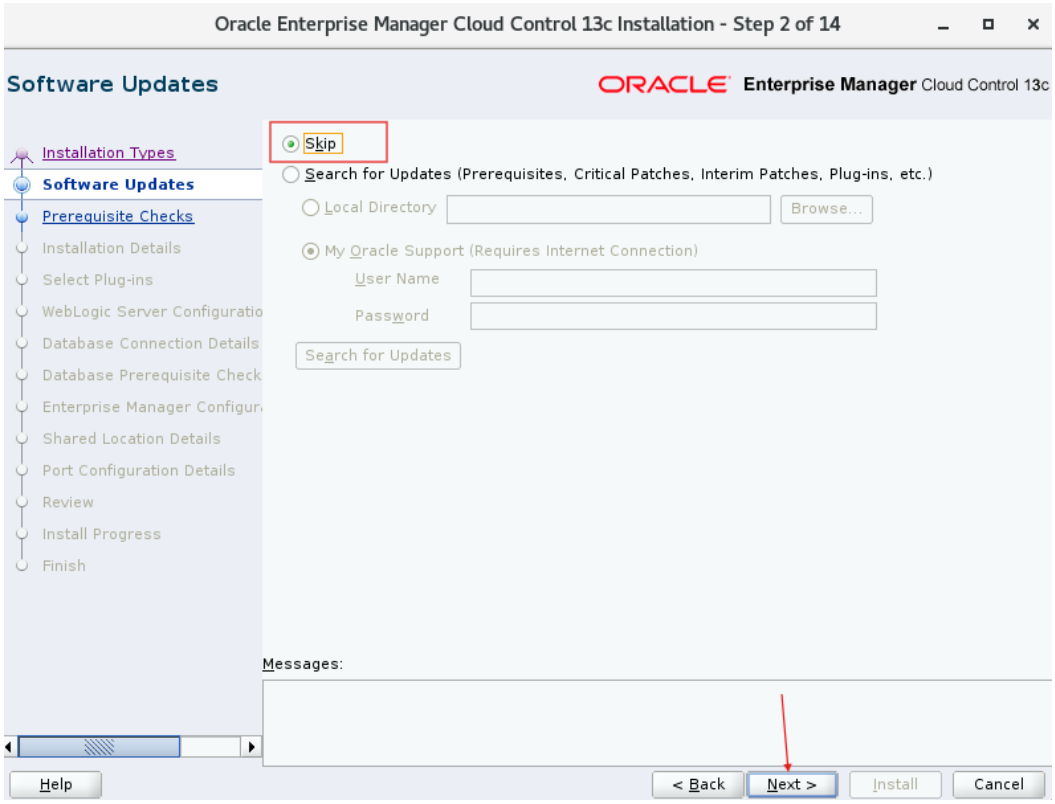
```
[oracle@OEM swlib]$ ./em13500_linux64.bin
Launcher log file is /tmp/OraInstall2026-09-14_01-36-26PM/launcher2026-09-14_01-36-26PM.log.
Extracting the installer . . . . . .
. . . . . Done
Checking monitor: must be configured to display at least 256 colors.    Actual 16777216    Passed
Checking swap space: must be greater than 512 MB.   Actual 33379 MB    Passed
Checking if this platform requires a 64-bit JVM.   Actual 64    Passed (64-bit not required)
Preparing to launch the Oracle Universal Installer from /tmp/OraInstall2026-09-14_01-36-26PM
ScratchPathValue :/tmp/OraInstall2026-09-14_01-36-26PM
Sep 14, 2026 1:37:34 PM org.apache.sshd.common.io.DefaultIoServiceFactoryFactory getIoServiceProvider
INFO: No detected/configured IoServiceFactoryFactory using Nio2ServiceFactoryFactory
```

The installer extracts the required installation components and launches the Oracle Universal Installer graphical interface.

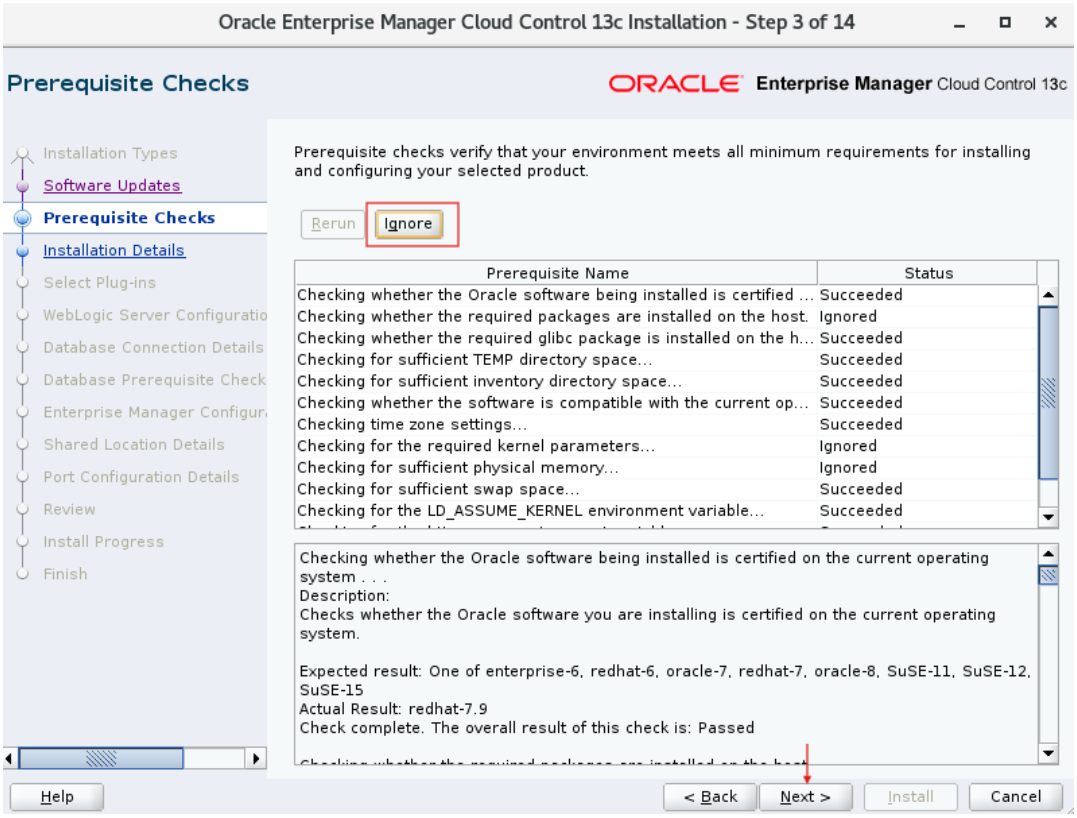
Choose **Create a new Enterprise Manager system → Advanced Install**. Click **Next**.



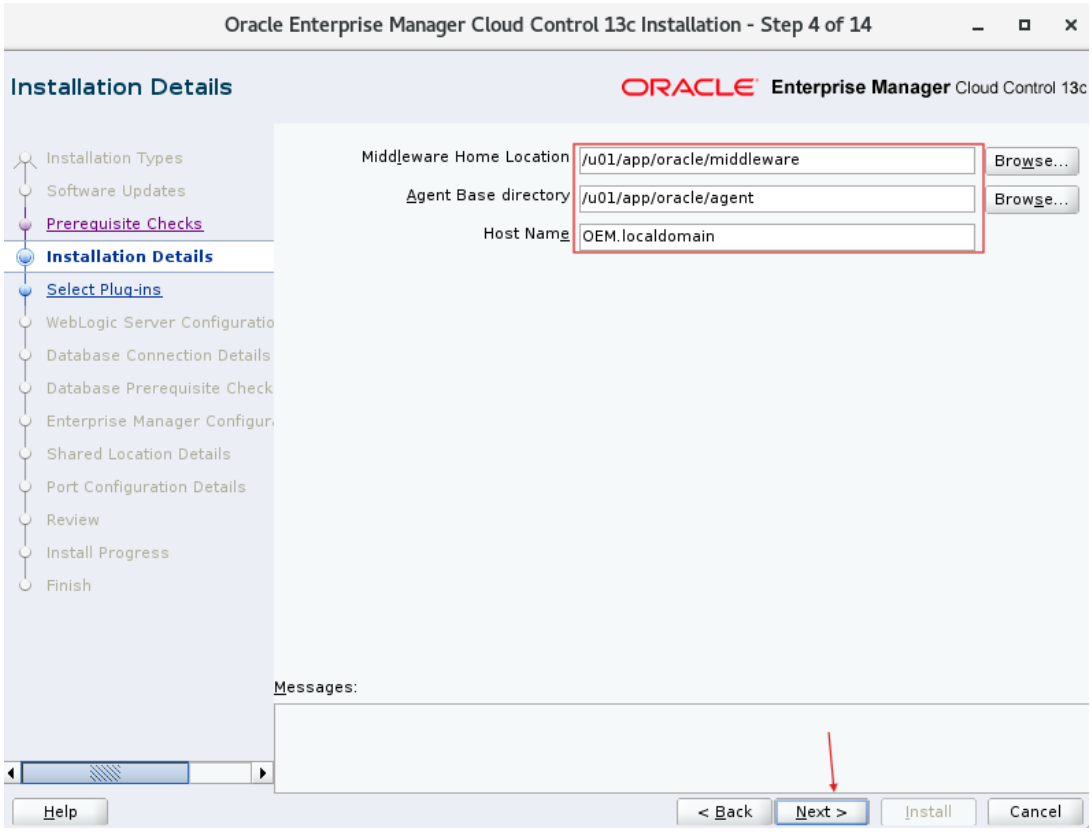
Select **Skip**. Click **Next**.



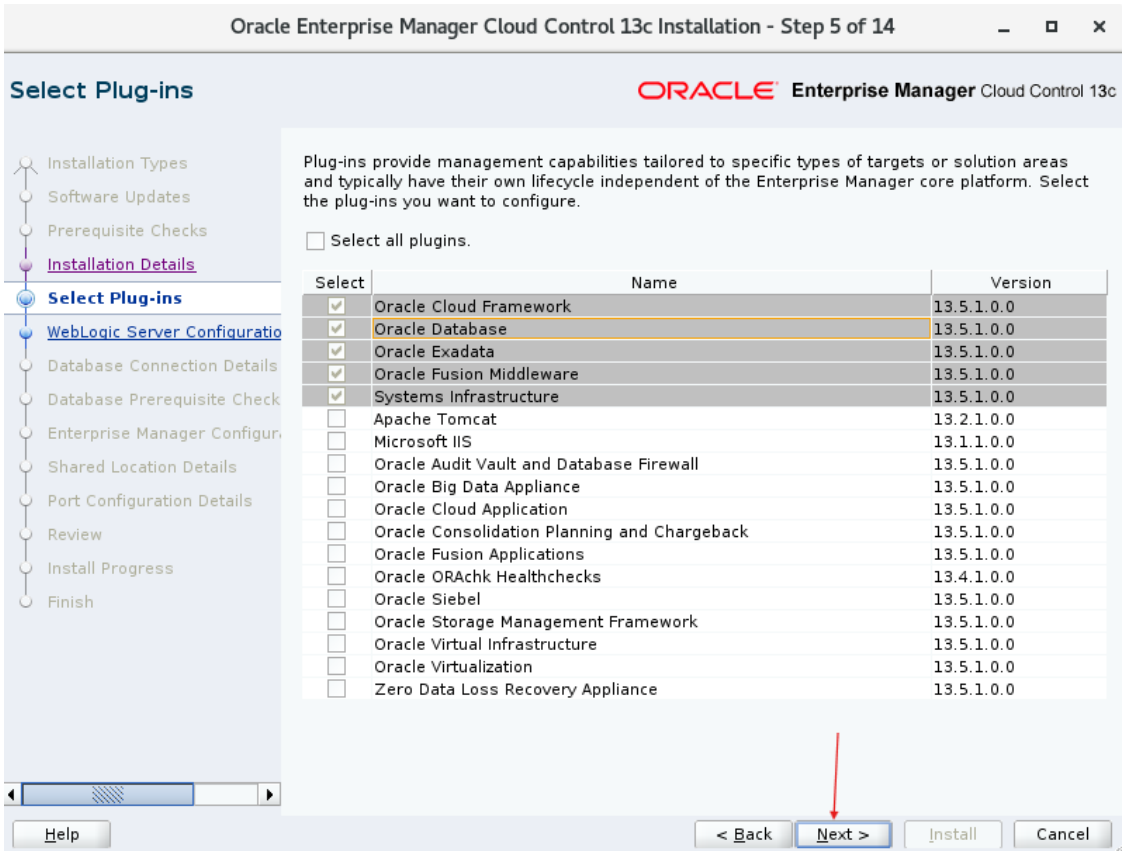
Installer verifies OS certification, packages, memory, swap. If any items show **Ignored/Failed** for non-critical items, select them and click **Ignore** if safe to proceed. Click **Next**.



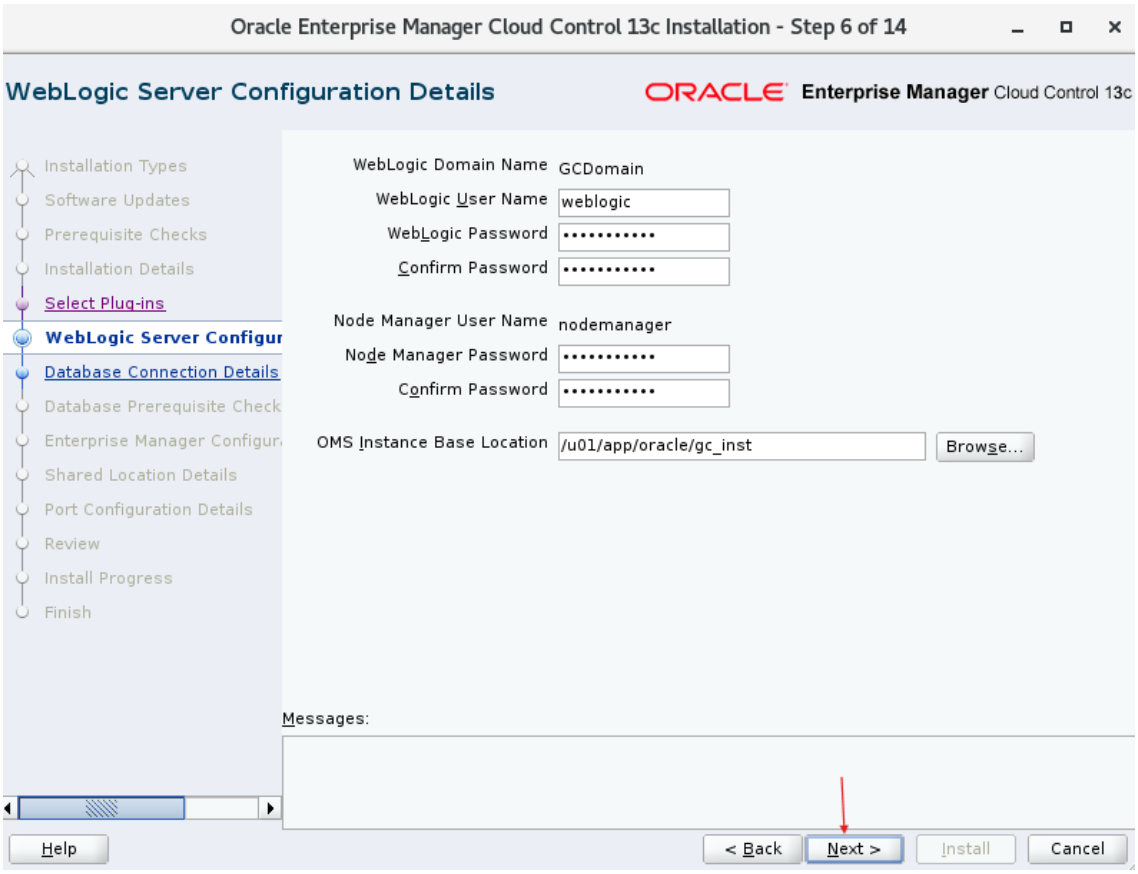
Give Middleware Home Location; Agent Base directory; & Host Name. Click **Next**.



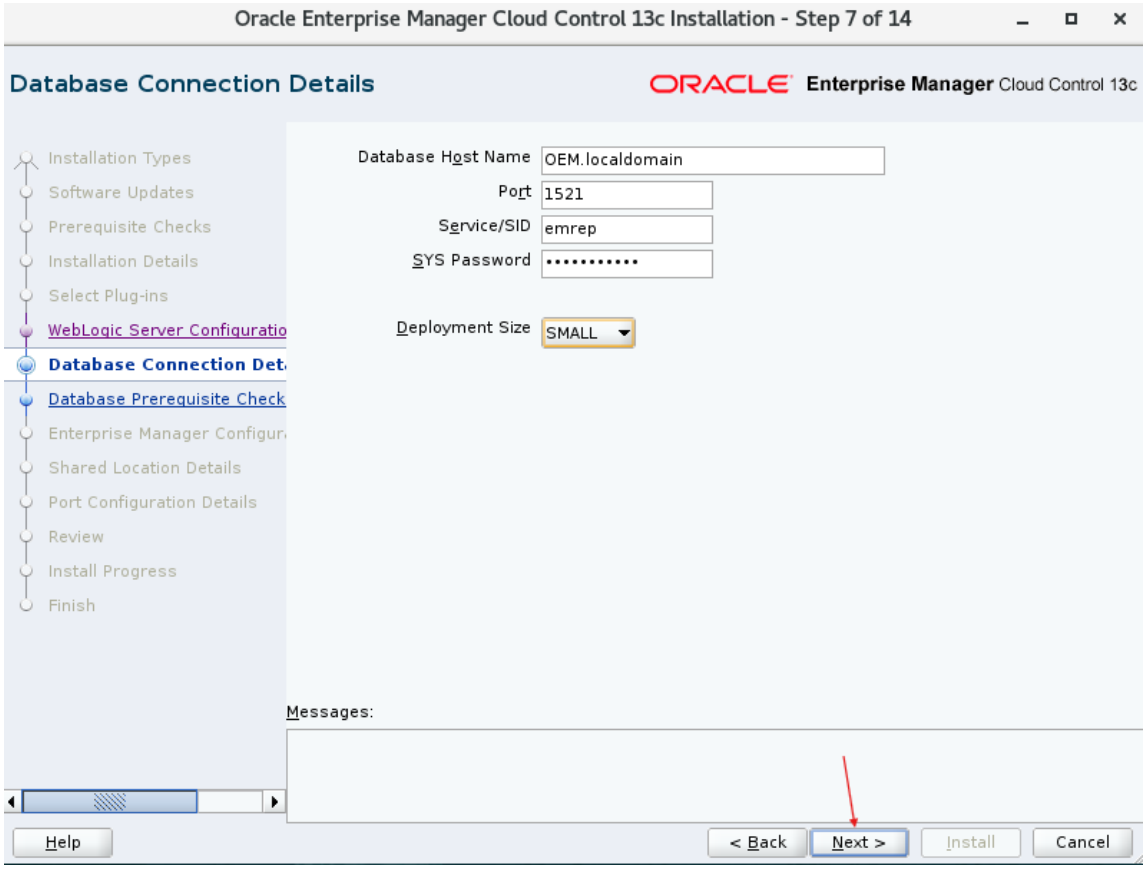
Leave the default checked plug-ins and Click **Next**.



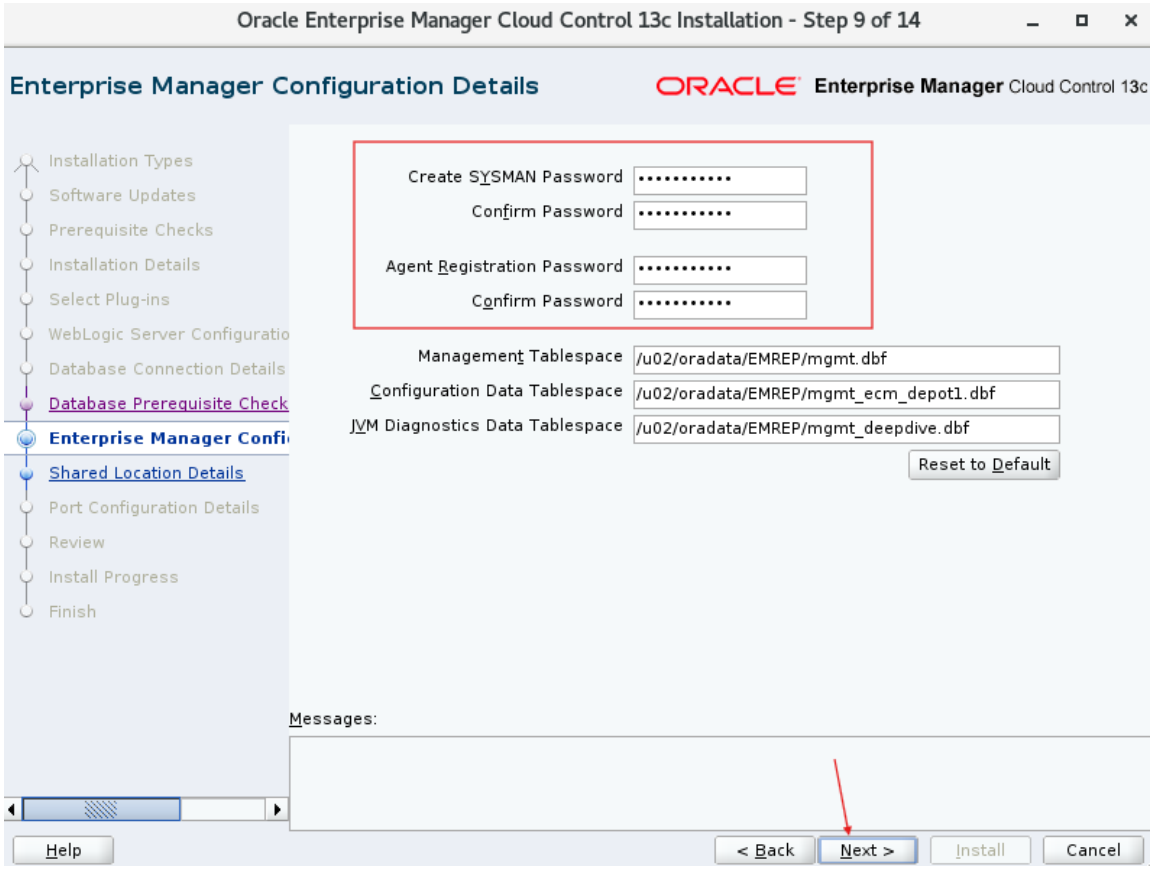
Give Domain Name; WebLogic User + password; Node Manager User + password; OMS Instance Base Location. Click **Next**.



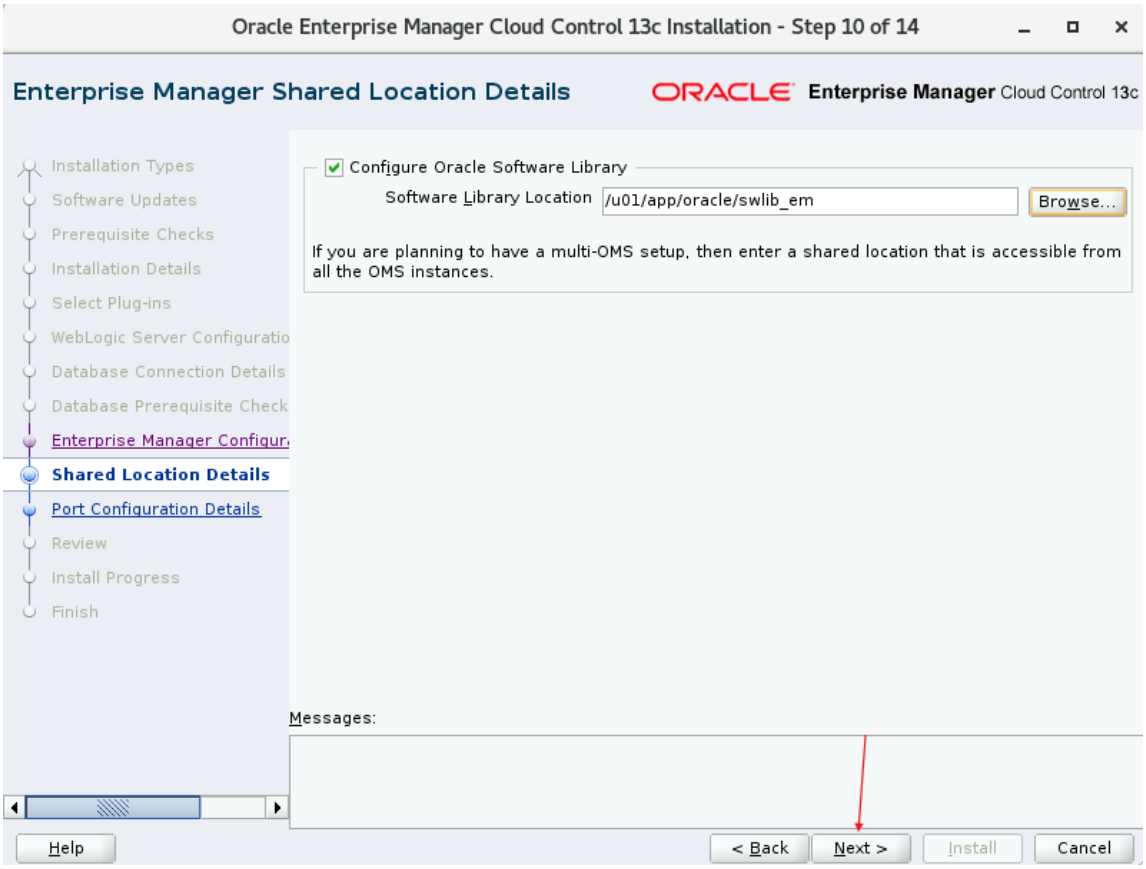
Fill the Database Host Name; Port; Service/SID; SYS Password: (from Phase 3); Deployment Size: **SMALL** (appropriate for this host's RAM). Click **Next**.



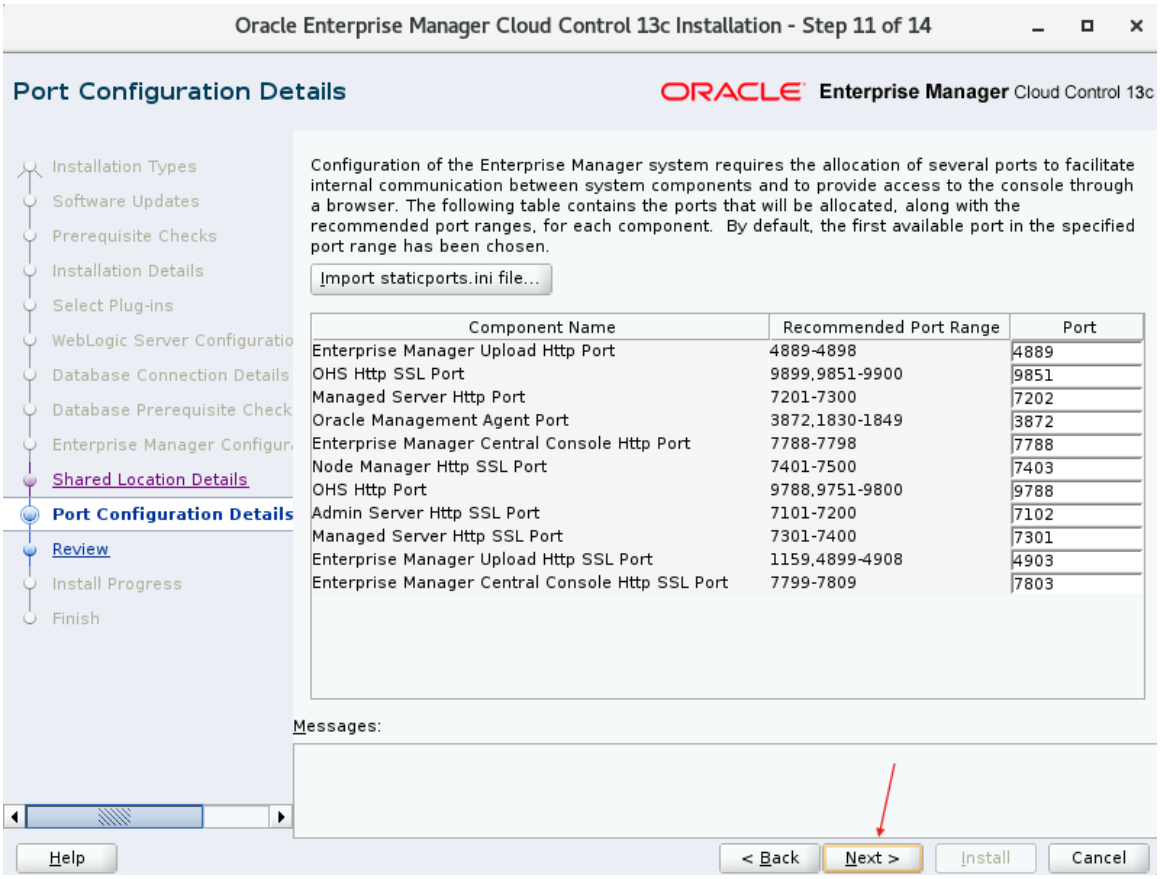
Create SYSMAN Password (this becomes your main console login) and Agent Registration Password. Confirm tablespace paths. Click **Next**.



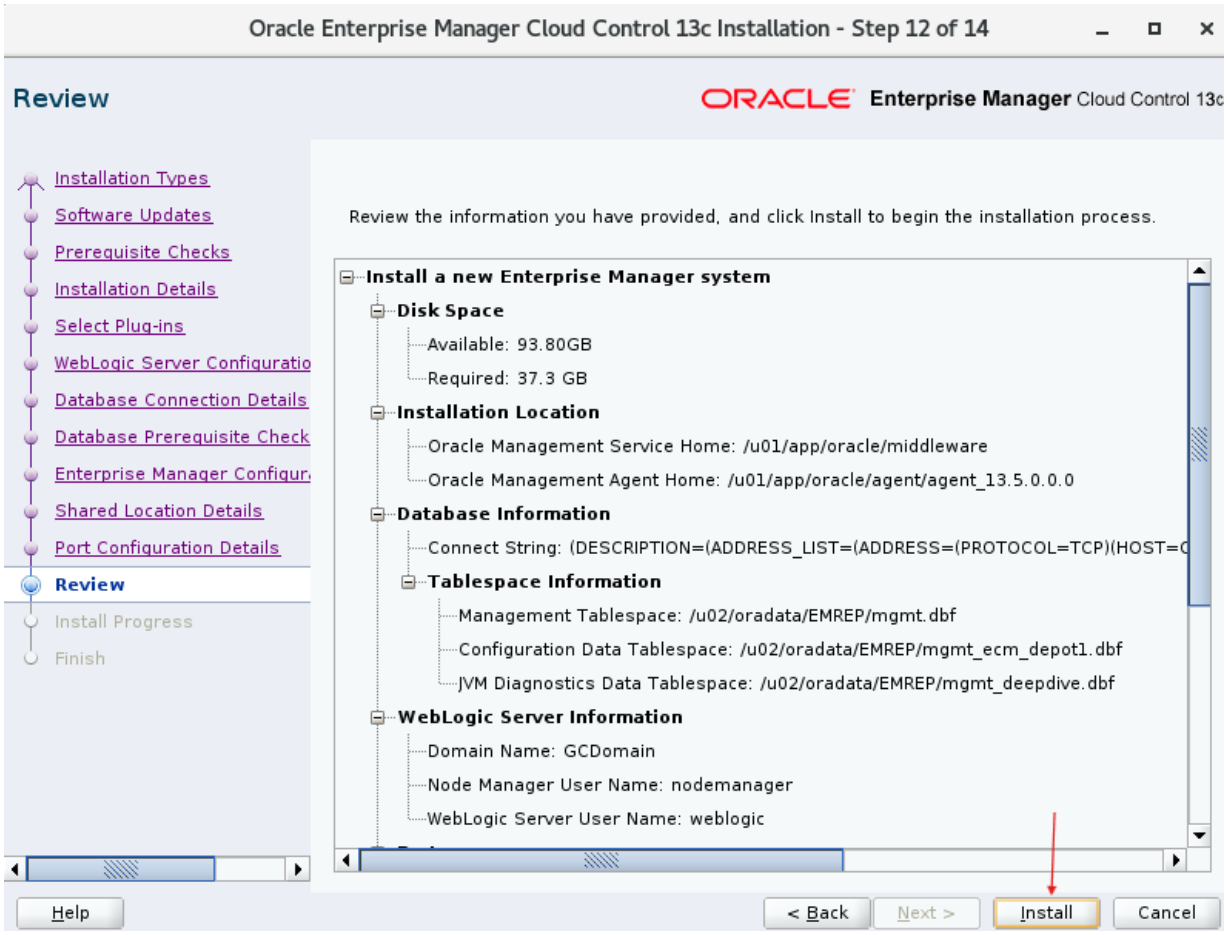
Software Library Location: use a dedicated **empty** directory (must NOT be the same folder holding the installer files, or validation fails). Click **Next**.

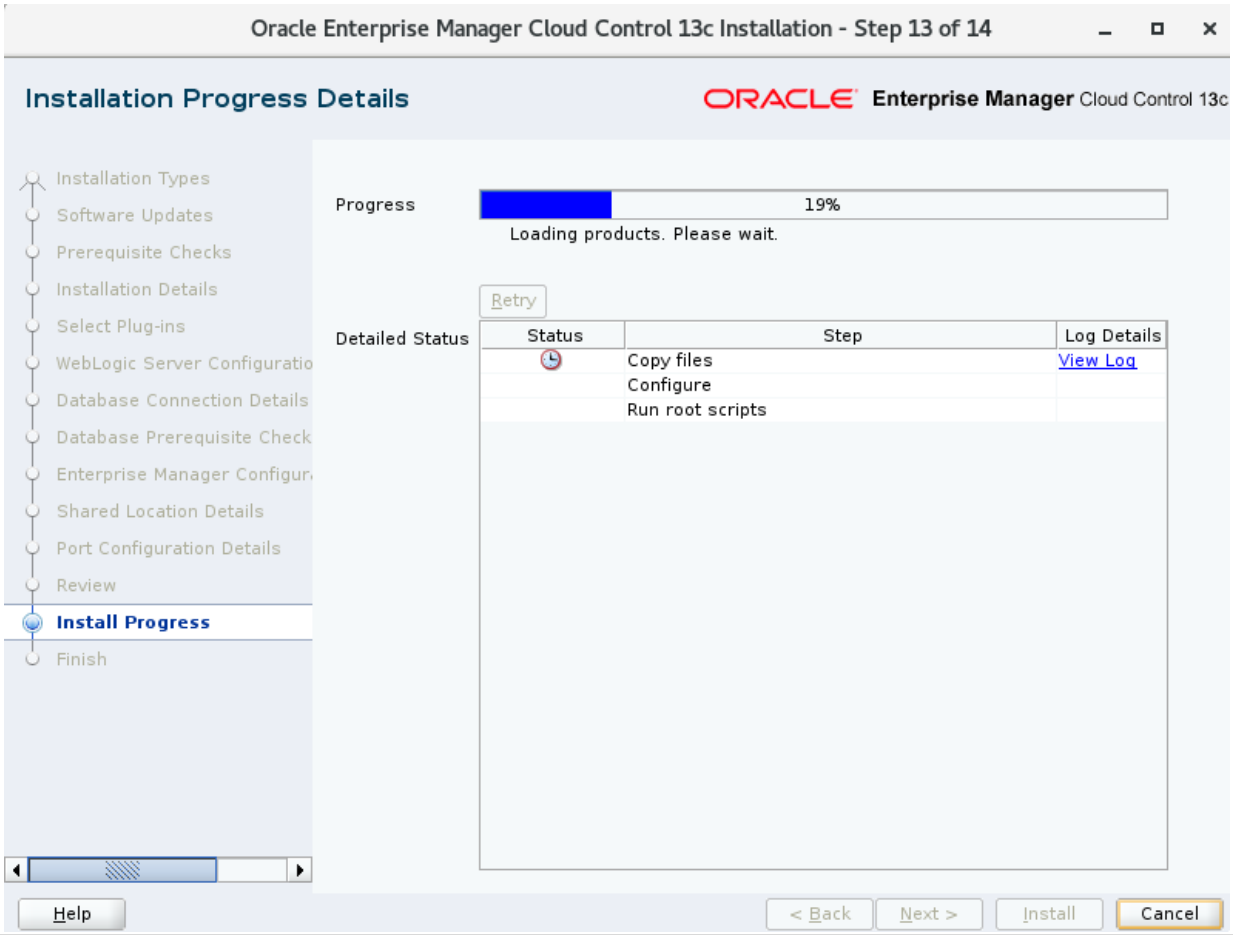


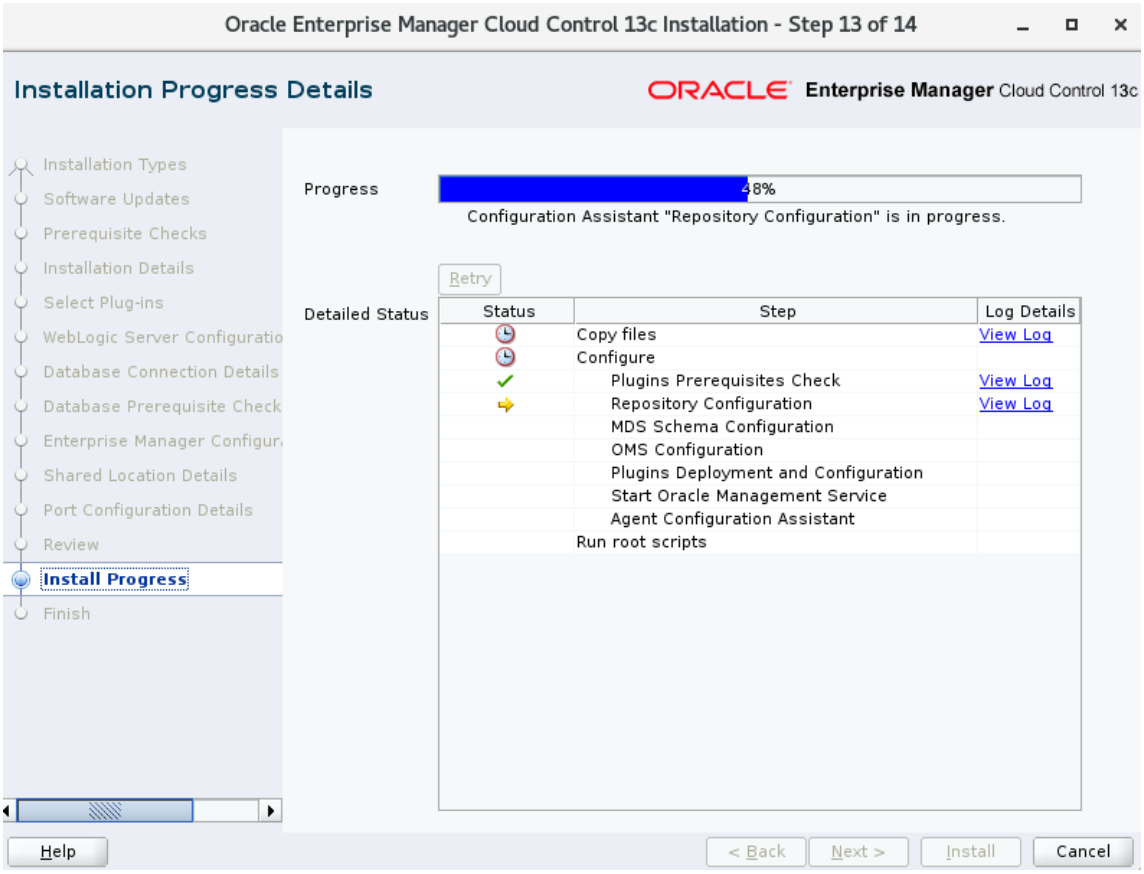
Review the auto-assigned ports, if defaults are fine. Click **Next**.

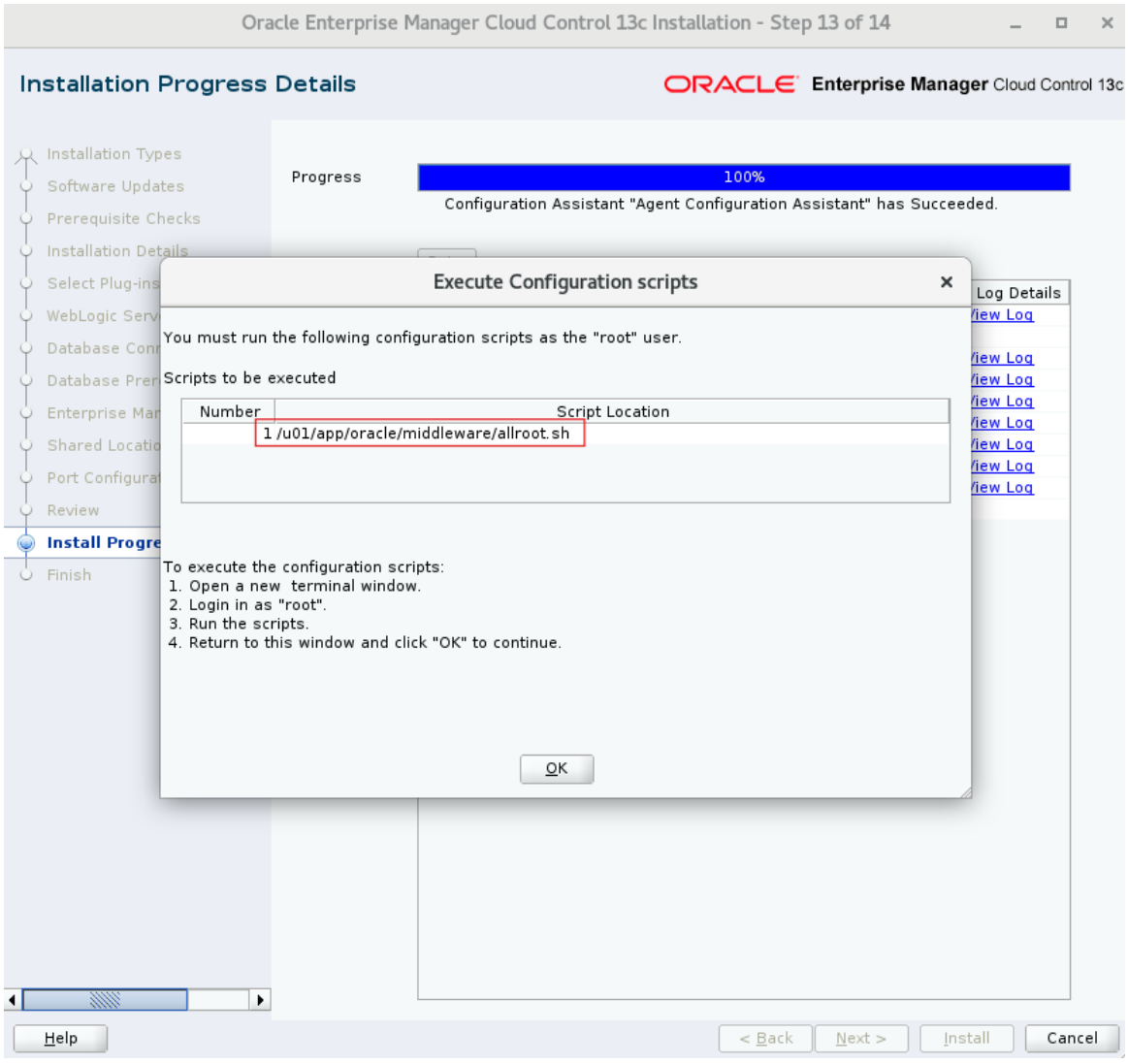


Confirms disk space, install locations, DB info, WebLogic domain info etc.. Click **Install**.






Oracle Enterprise Manager Cloud Control 13c Installation - Step 13 of 14
Installation Progress Details
ORACLE Enterprise Manager Cloud Control 13c
Installation Types
Software Updates
Prerequisite Checks
Installation Details
Select Plug-ins
WebLogic Server Configuratio
Database Connection Details
Database Prerequisite Check
Enterprise Manager Configur
Shared Location Details
Port Configuration Details
Review
Install Progress
Finish
Progress
48%
Configuration Assistant "Repository Configuration" is in progress.
Retry
Detailed Status
Status
Step
Log Details
Copy files
View Log
Configure
Plugins Prerequisites Check
View Log
Repository Configuration
View Log
MDS Schema Configuration
OMS Configuration
Plugins Deployment and Configuration
Start Oracle Management Service
Agent Configuration Assistant
Run root scripts
Help
< Back
Next >
Install
Cancel


Oracle Enterprise Manager Cloud Control 13c Installation - Step 13 of 14
Installation Progress Details
ORACLE Enterprise Manager Cloud Control 13c
Installation Types
Software Updates
Prerequisite Checks
Installation Details
Select Plug-ins
WebLogic Serv
Database Con
Database Prer
Enterprise Mar
Shared Locatio
Port Configura
Review
Install Progre
Finish
Progress
100%
Configuration Assistant "Agent Configuration Assistant" has Succeeded.
Log Details
View Log
Execute Configuration scripts
You must run the following configuration scripts as the "root" user.
Scripts to be executed
Number
Script Location
1 /u01/app/oracle/middleware/allroot.sh
To execute the configuration scripts:
1. Open a new terminal window.
2. Login in as "root".
3. Run the scripts.
4. Return to this window and click "OK" to continue.
OK
Help
< Back
Next >
Install
Cancel

"Execute Configuration scripts" appears listing allroot.sh : open a root terminal and run it.

```
[root@OEM ~]# /u01/app/oracle/middleware/allroot.sh

Starting to execute allroot.sh .........

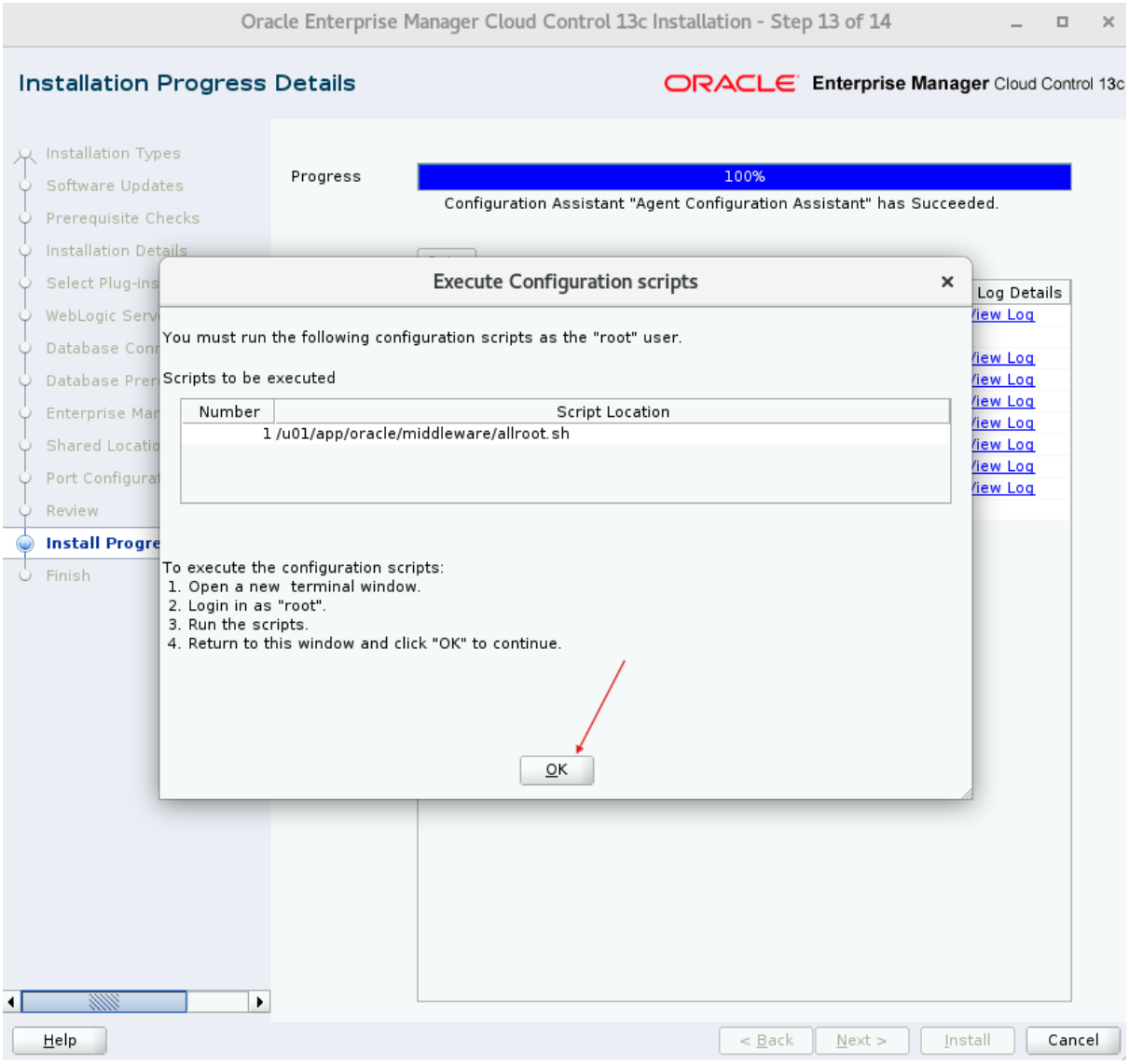
Starting to execute /u01/app/oracle/middleware/root.sh ......
Check /u01/app/oracle/middleware/install/root_OEM.localdomain_2026-09-14_18-15-14.log for the output of
root script

Finished product-specific root actions.
/etc exist
Finished execution of  /u01/app/oracle/middleware/root.sh ......

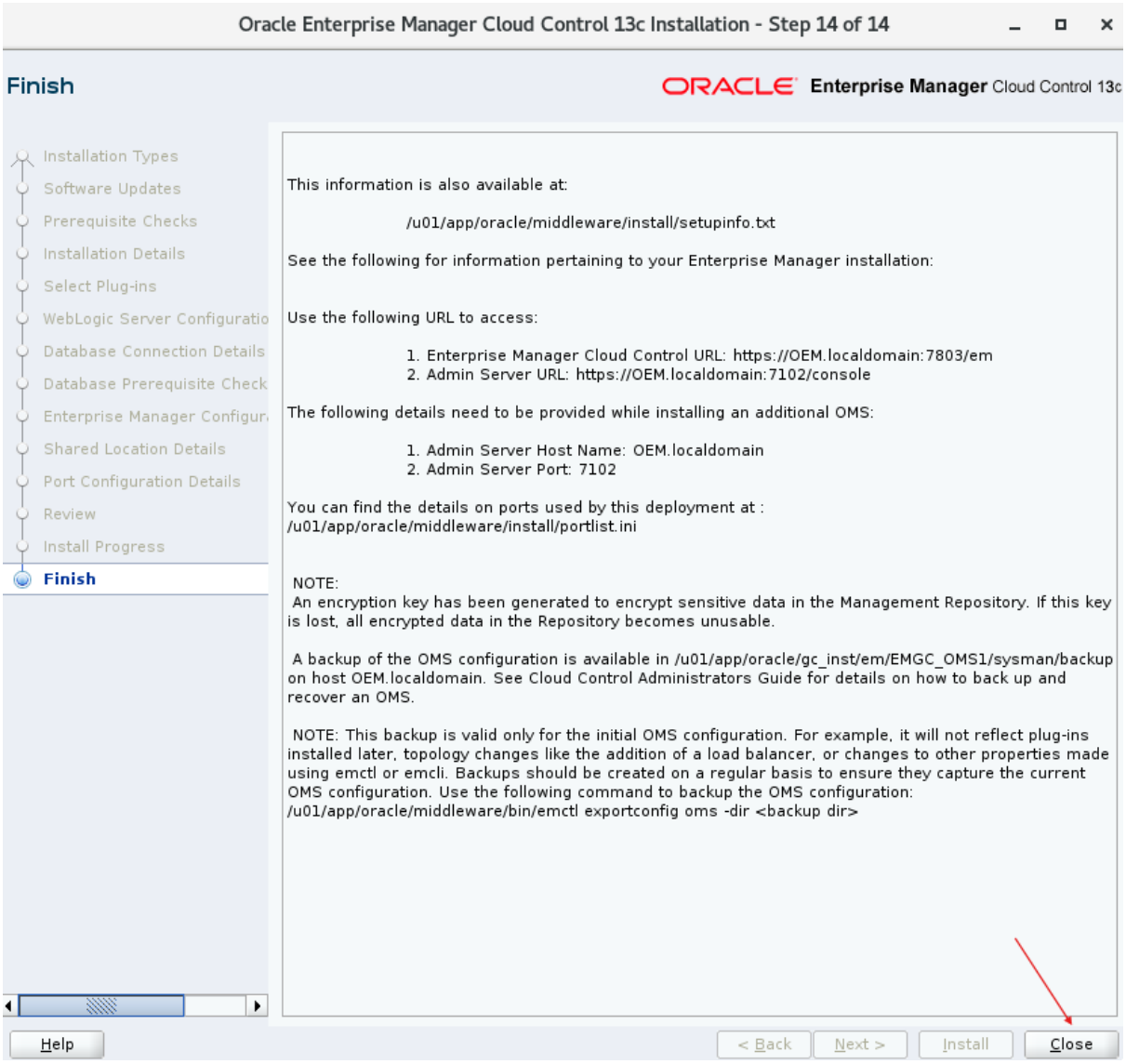

Starting to execute /u01/app/oracle/agent/agent_13.5.0.0.0/root.sh ......

Finished product-specific root actions.
/etc exist
Finished execution of  /u01/app/oracle/agent/agent_13.5.0.0.0/root.sh ......
[root@OEM ~]#
```

Once root script run successfully then click **ok**.



Displays the EM Console URL (https://OEM.localdomain:7803/em), Admin Server URL, and important notes about the EM encryption key backup location. Click **Close**.



PHASE 6 : Configure SSH Connectivity Between OEM and RAC Nodes

Objective : SSH connectivity from the OEM server to both RAC nodes is required for the Management Agent deployment workflow and remote administration.

The implementation configured SSH key-based authentication from the OEM server to:

node1 & node2

Generate SSH Keys

On the OEM server, as the Oracle user:

```
[oracle@OEM ~]$ ssh-keygen -t rsa -N "" -f ~/.ssh/id_rsa
Generating public/private rsa key pair.
Your identification has been saved in /home/oracle/.ssh/id_rsa.
Your public key has been saved in /home/oracle/.ssh/id_rsa.pub.
The key fingerprint is:
```

```
SHA256:bssX6ZaXGNlPxgs7GhVmmS1mGoBoP0/0Ni0VyMRmQmA oracle@OEM.localdomain
The key's randomart image is:
+---[RSA 2048]----+
|     .E+o+....   |
|    o.. o.* .+   |
|   . . . =.oX .  |
|      o . =B.o   |
|       +S..*..   |
|       .. =.o +  |
|        o..= B . |
|       o .*.= o  |
|        oo.o .   |
+----[SHA256]-----+
[oracle@OEM ~]$
[oracle@OEM ~]$ ls -la ~/.ssh/
total 12
drwx------.  2 oracle oinstall   38 Sep 15 12:39 .
drwx------. 20 oracle oinstall 4096 Sep 15 12:39 ..
-rw-------.  1 oracle oinstall 1679 Sep 15 12:39 id_rsa
-rw-r--r--.  1 oracle oinstall  404 Sep 15 12:39 id_rsa.pub
[oracle@OEM ~]$
```

Copy the Public Key to Node 1 & Test SSH Connectivity

```
[oracle@OEM ~]$ ssh-copy-id oracle@node1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/oracle/.ssh/id_rsa.pub"
The authenticity of host 'node1 (192.168.18.11)' can't be established.
ECDSA key fingerprint is SHA256:5vzPnNy3ocCY/HC468HoD5PR0bonU0CfCeKJDHfuMoI.
ECDSA key fingerprint is MD5:e8:de:d8:54:c5:ed:c3:1b:f7:fc:92:83:cb:a4:02:12.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install
the new keys
oracle@node1's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'oracle@node1'"
and check to make sure that only the key(s) you wanted were added.

[oracle@OEM ~]$
[oracle@OEM ~]$ ssh node1
Last login: Tue Sep 15 10:03:24 2026
[oracle@node1 ~]$ exit
logout
Connection to node1 closed.
[oracle@OEM ~]$
```

This adds the OEM server's public key to the authorized keys of the Oracle user on node1. Confirm that the Oracle user can connect successfully without being asked for the Oracle password each time.

Copy the Public Key to Node 2 & Test SSH Connectivity

```
[oracle@OEM ~]$ ssh-copy-id oracle@node2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/oracle/.ssh/id_rsa.pub"
The authenticity of host 'node2 (192.168.18.12)' can't be established.
ECDSA key fingerprint is SHA256:5vzPnNy3ocCY/HC468HoD5PR0bonU0CfCeKJDHfuMoI.
ECDSA key fingerprint is MD5:e8:de:d8:54:c5:ed:c3:1b:f7:fc:92:83:cb:a4:02:12.
Are you sure you want to continue connecting (yes/no)? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install
```

```
the new keys
oracle@node2's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'oracle@node2'"
and check to make sure that only the key(s) you wanted were added.

[oracle@OEM ~]$
[oracle@OEM ~]$ ssh node2
Last login: Tue Sep 15 09:56:58 2026
[oracle@node2 ~]$ exit
logout
Connection to node2 closed.
[oracle@OEM ~]$
```

The implementation completed SSH key deployment to both RAC nodes.

PHASE 7 : Deploy Management Agents to RAC Nodes

Objective : Management Agents provide the communication layer between the RAC hosts and the OEM Management Service.

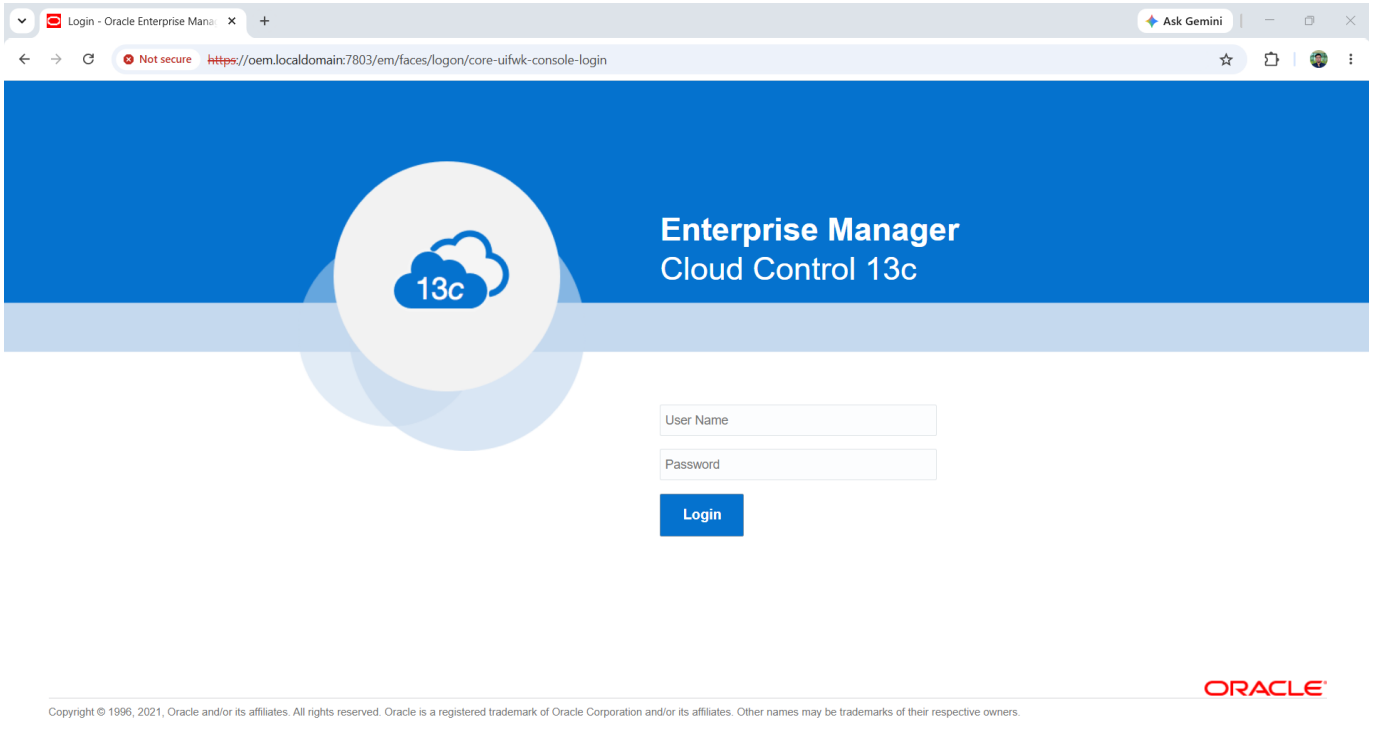
The implementation deployed Enterprise Manager 13.5 Management Agents to:

node1.localdomain & node2.localdomain

Log in to Enterprise Manager

Open the Enterprise Manager console in a web browser.

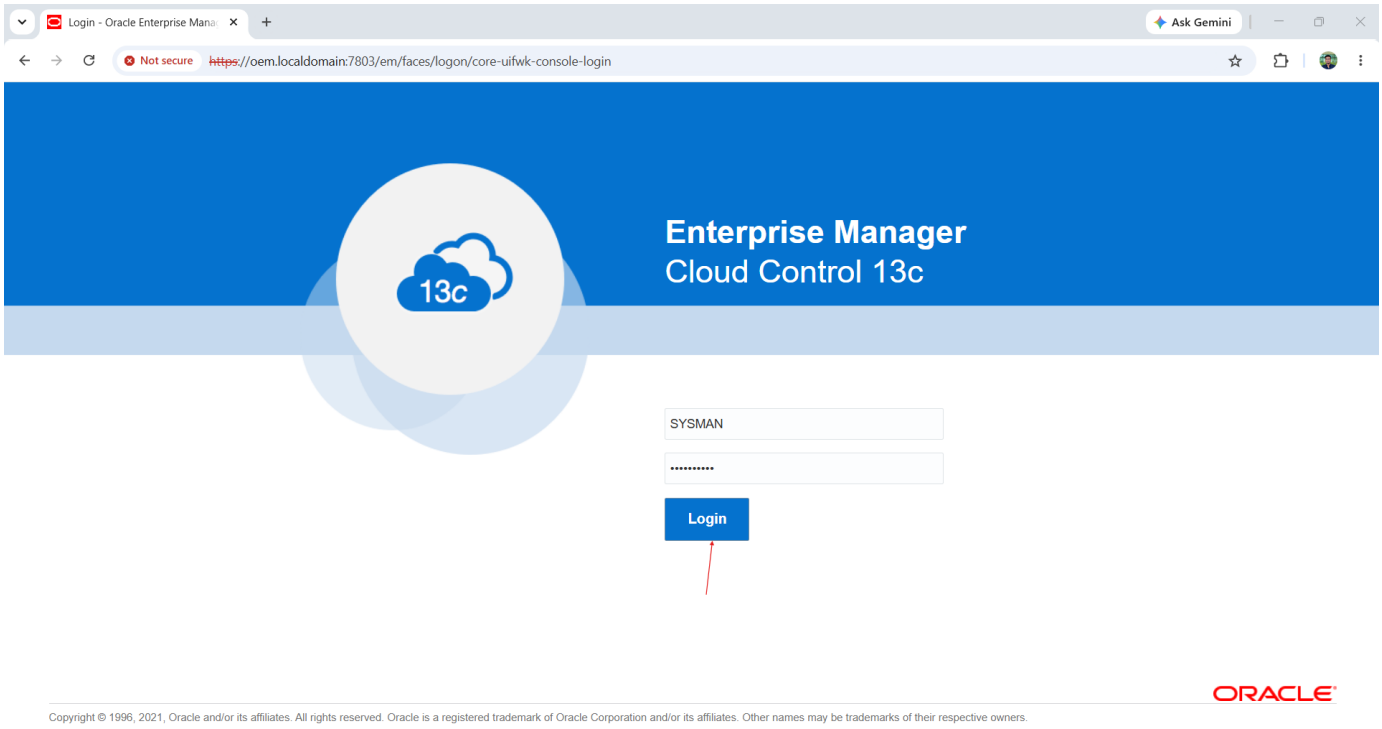
Use the Enterprise Manager console URL configured during installation.



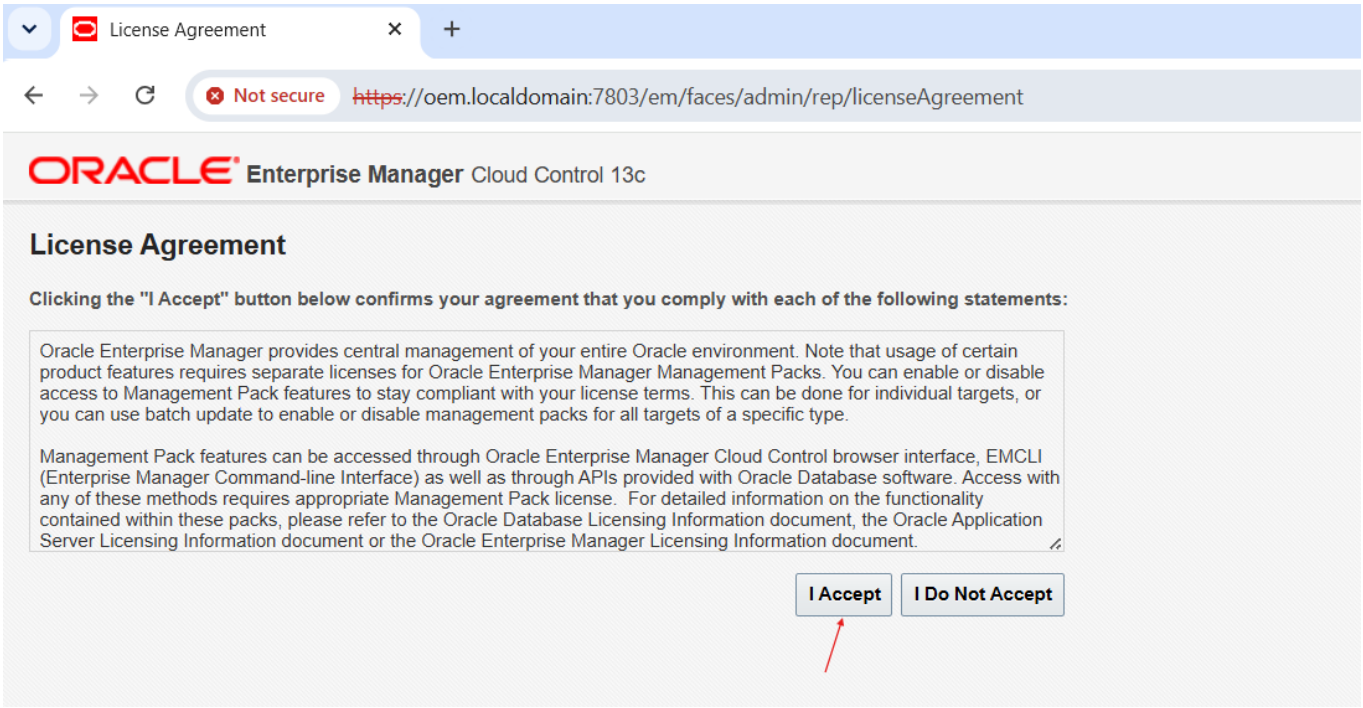
Log in using:

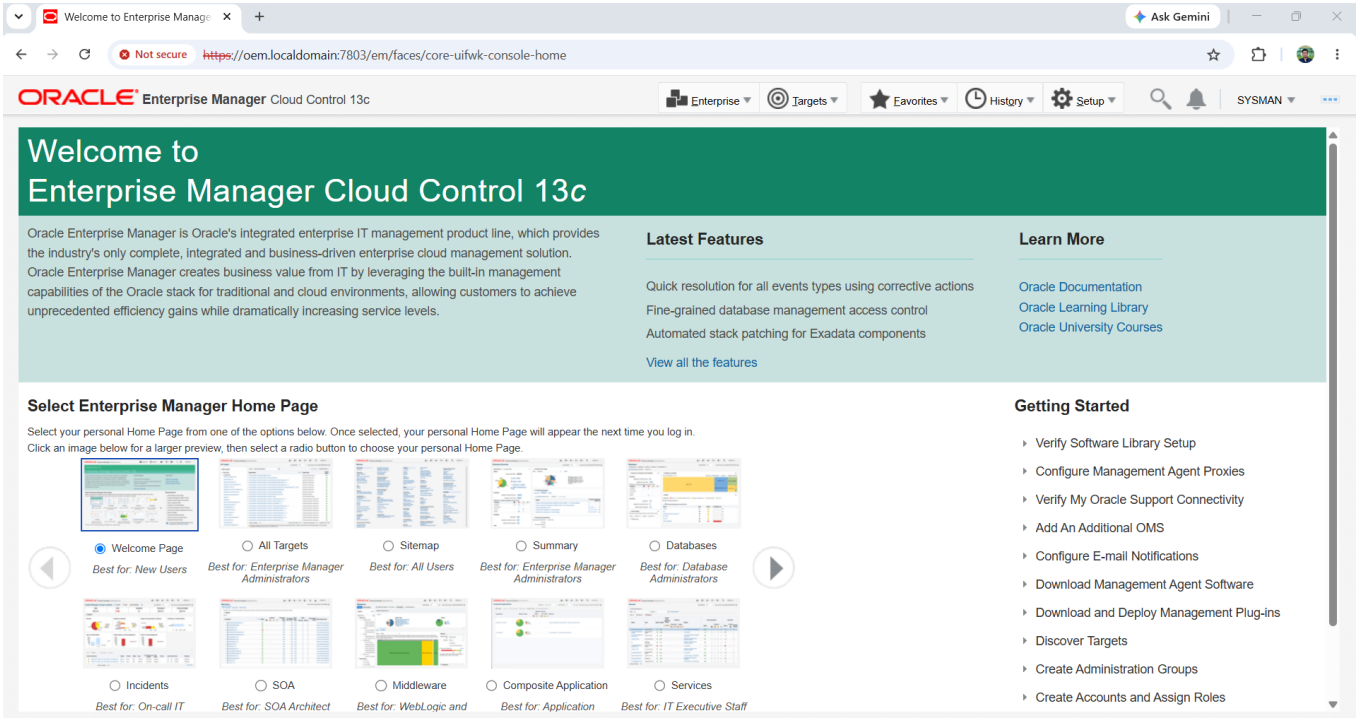
Username: SYSMAN

Password: <SYSMAN_PASSWORD>

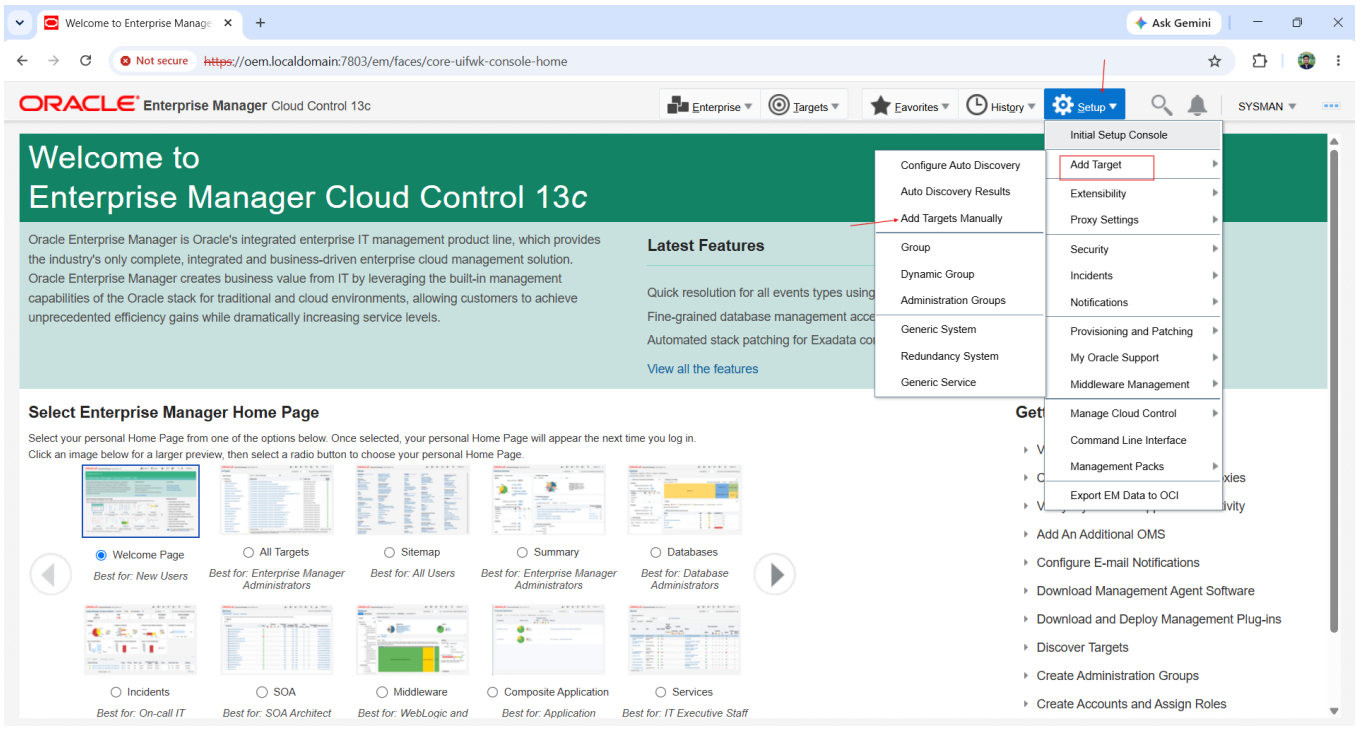


Accept the licence agreement.

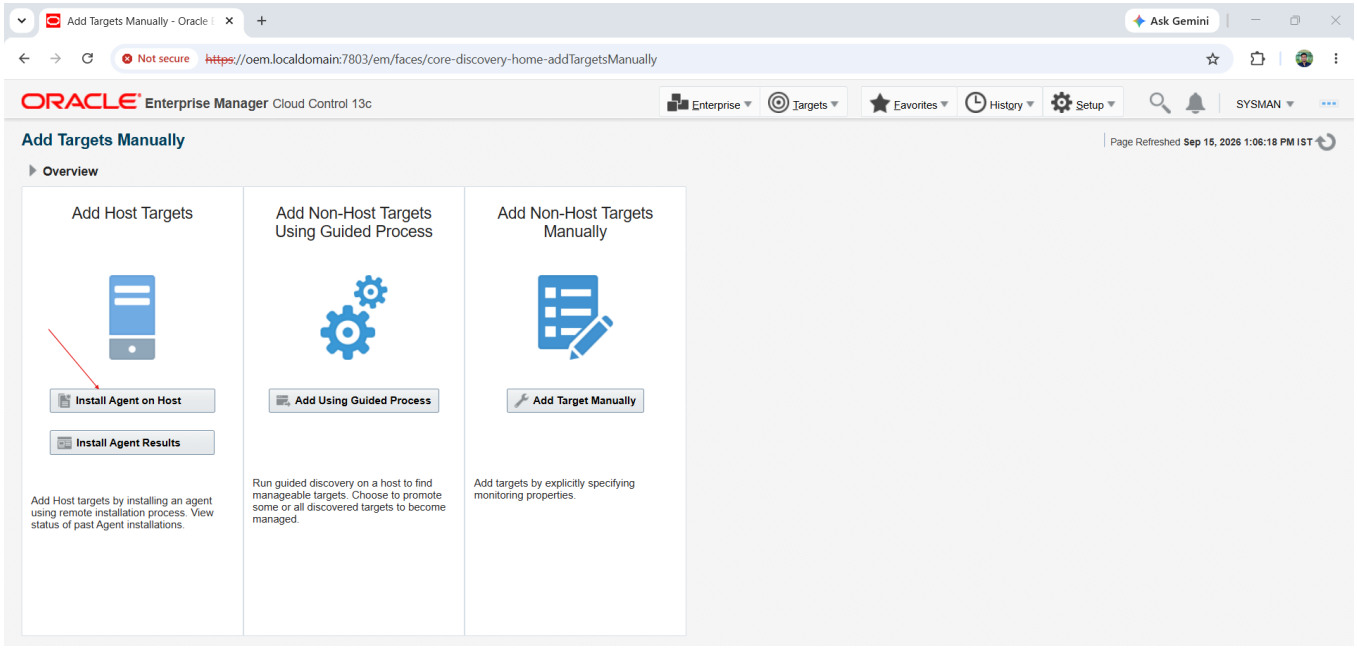




Navigate: **Setup (gear icon) → Add Target → Add Targets Manually → Install Agent on Host**.



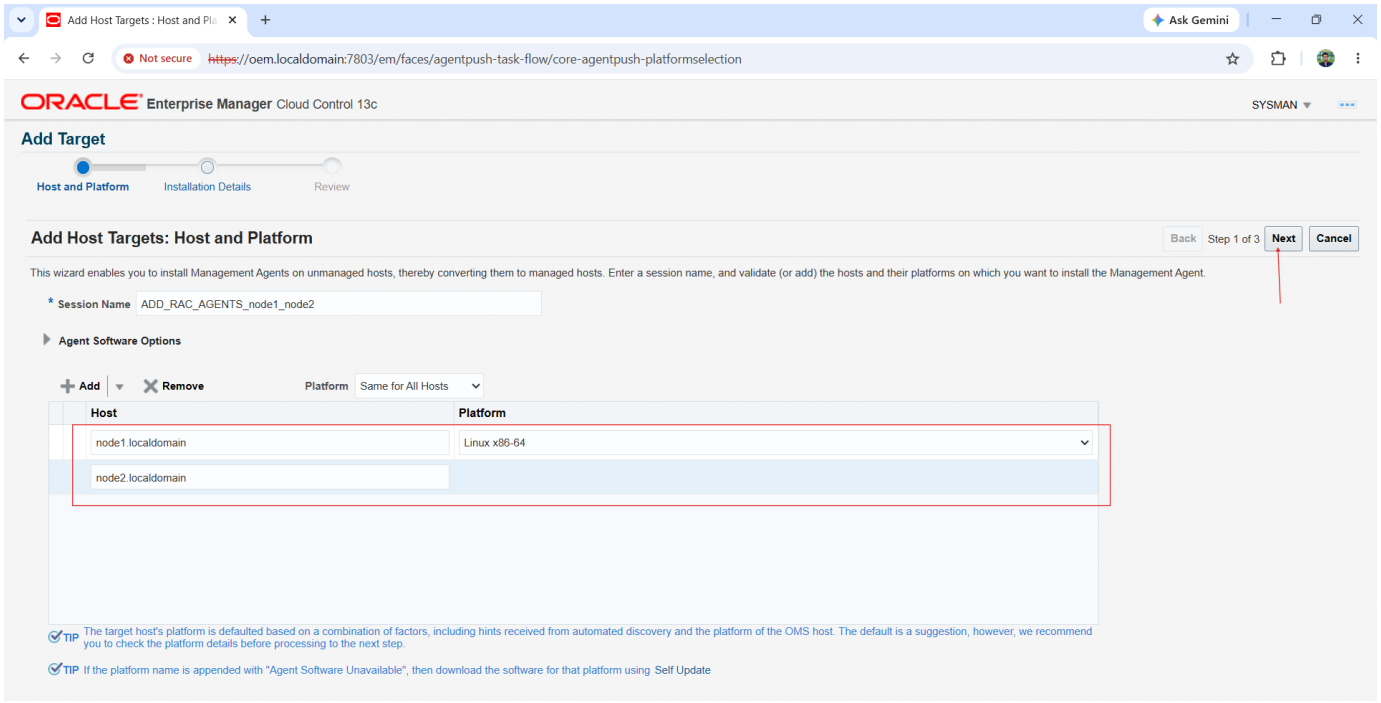
This opens the **Add Targets Manually** landing page with three options. click **Install Agent on Host** under "Add Host Targets."



Step 1 of 3 : Host and Platform

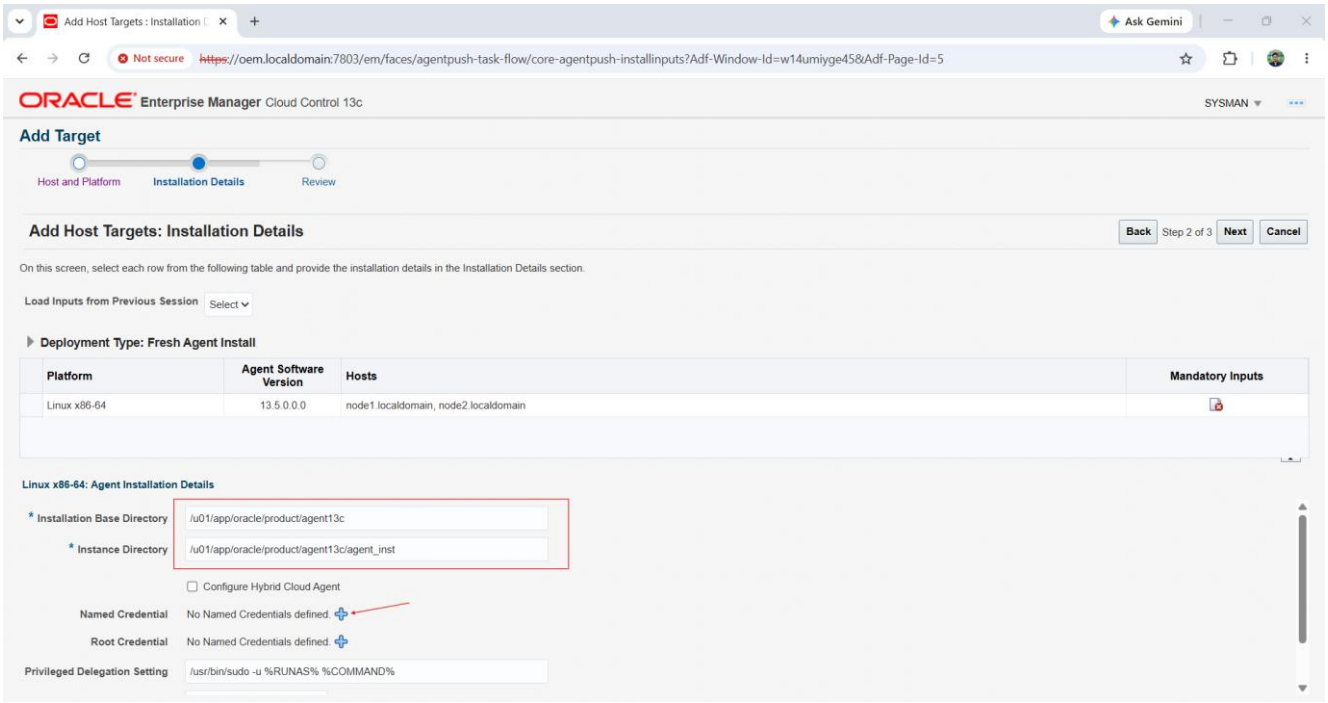
- **Session Name:** rename to something identifiable, e.g.

  ADD_RAC_AGENTS_node1_node2.
- **Add** two rows under the Host table: node1.localdomain and node2.localdomain.
- **Platform:** select **Linux x86-64** (set to "Same for All Hosts" to apply to both at once).
- Click **Next**.

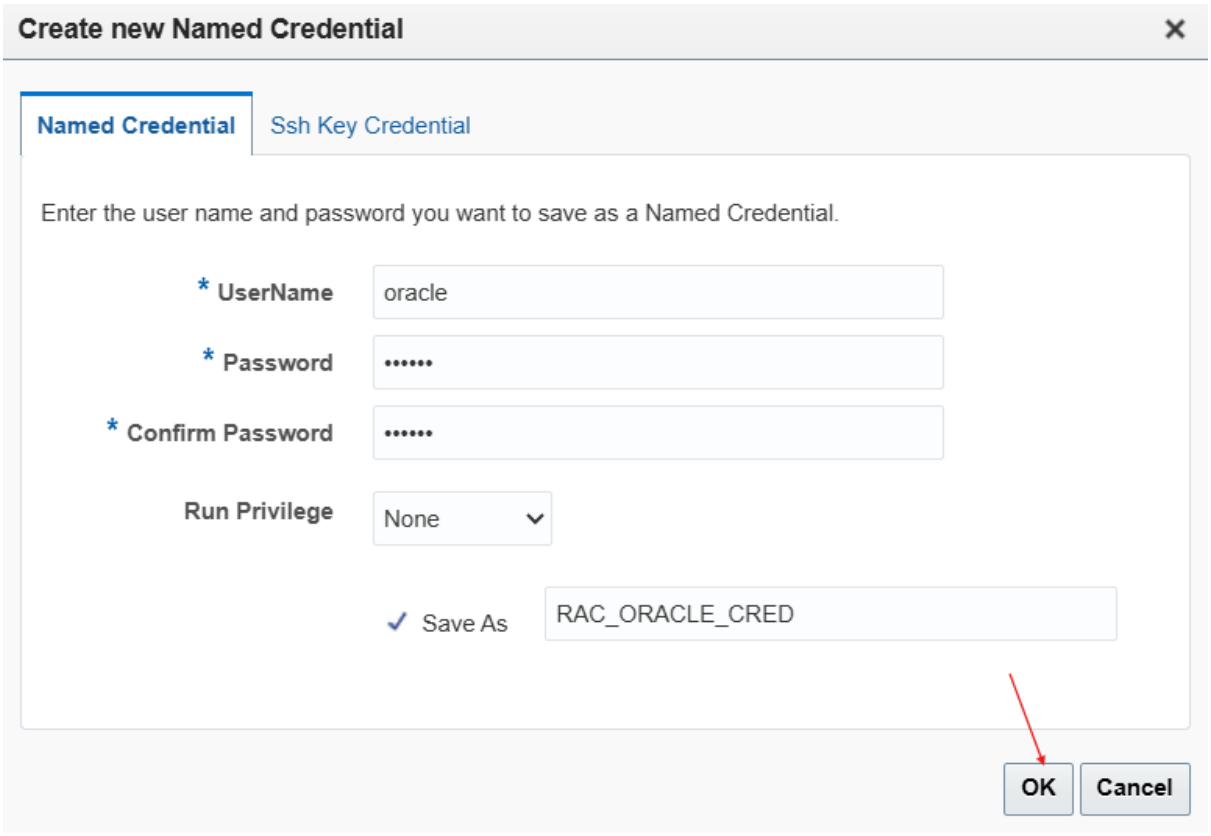


Step 2 of 3 : Installation Details

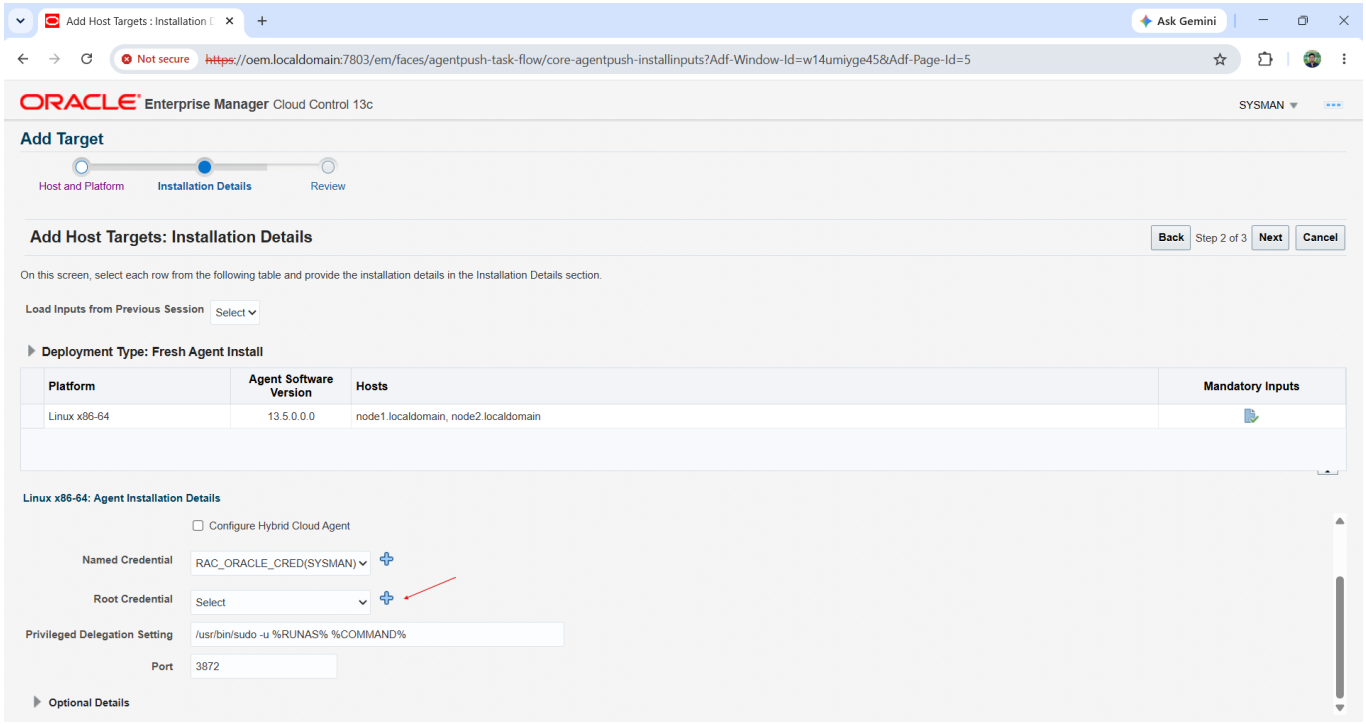
- **Installation Base Directory:** /u01/app/oracle/product/agent13c
- **Instance Directory:** /u01/app/oracle/product/agent13c/agent_inst

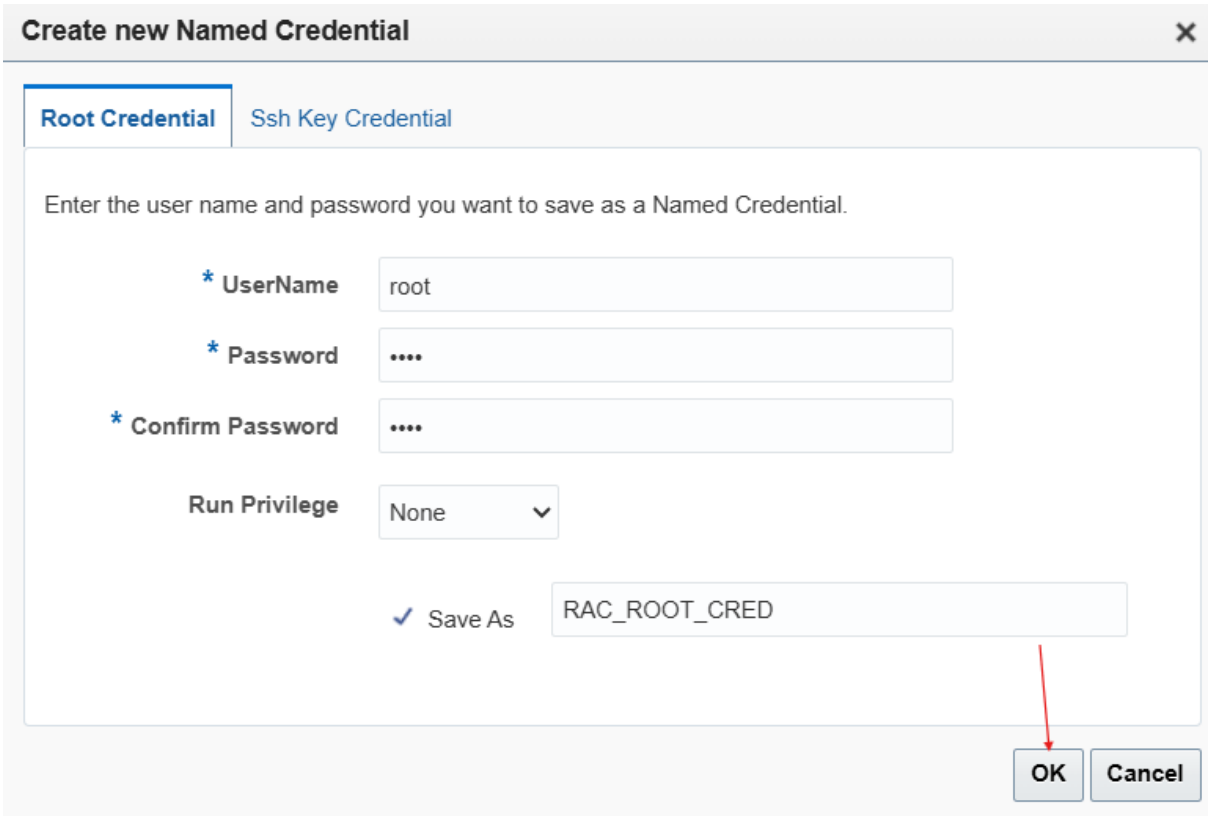


- **Named Credential:** click the + icon → enter Username oracle, Password, Run Privilege None, Save As RAC_ORACLE_CRED. Click **OK**.



- **Root Credential:** click the + icon → enter Username root, Password, Save As RAC_ROOT_CRED. Click **OK**.

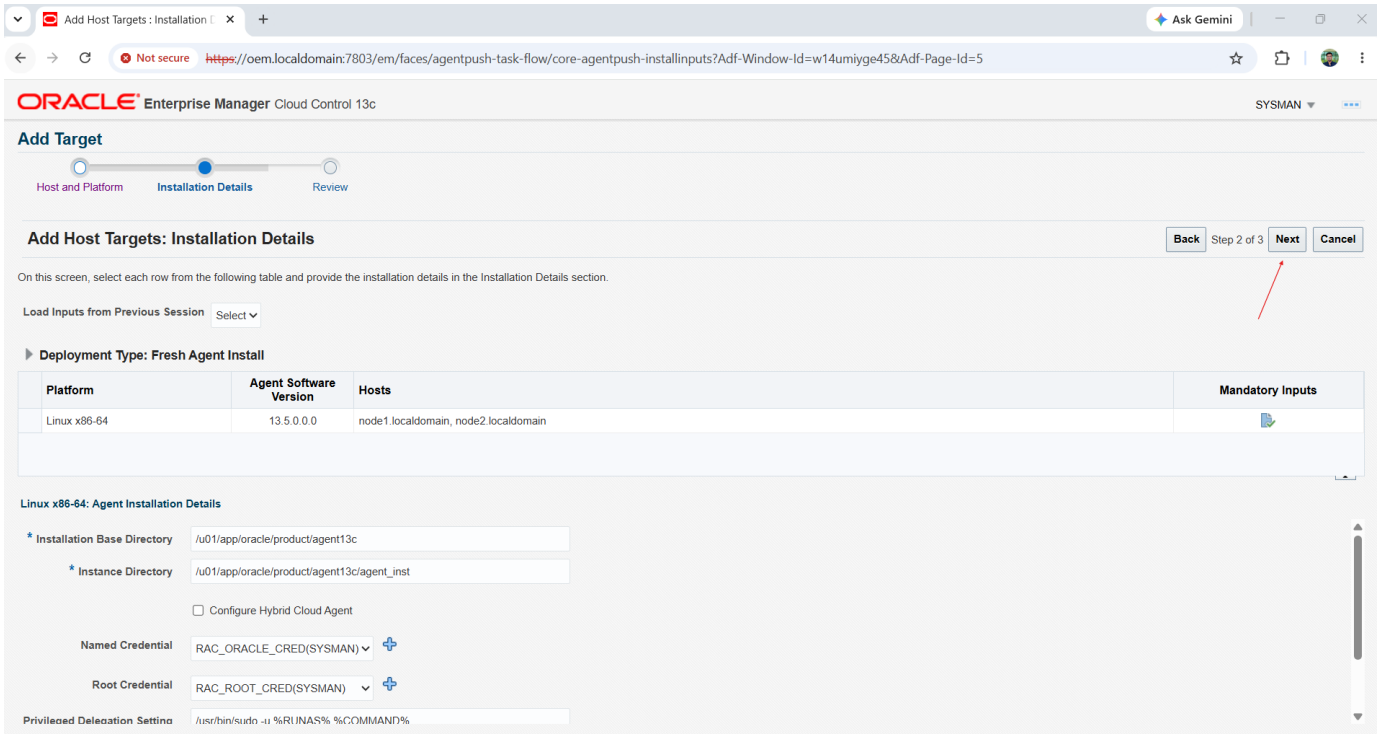




(Only needed if the oracle OS user doesn't have sudo rights on the RAC nodes, a direct root credential bypasses the sudo delegation setting.)

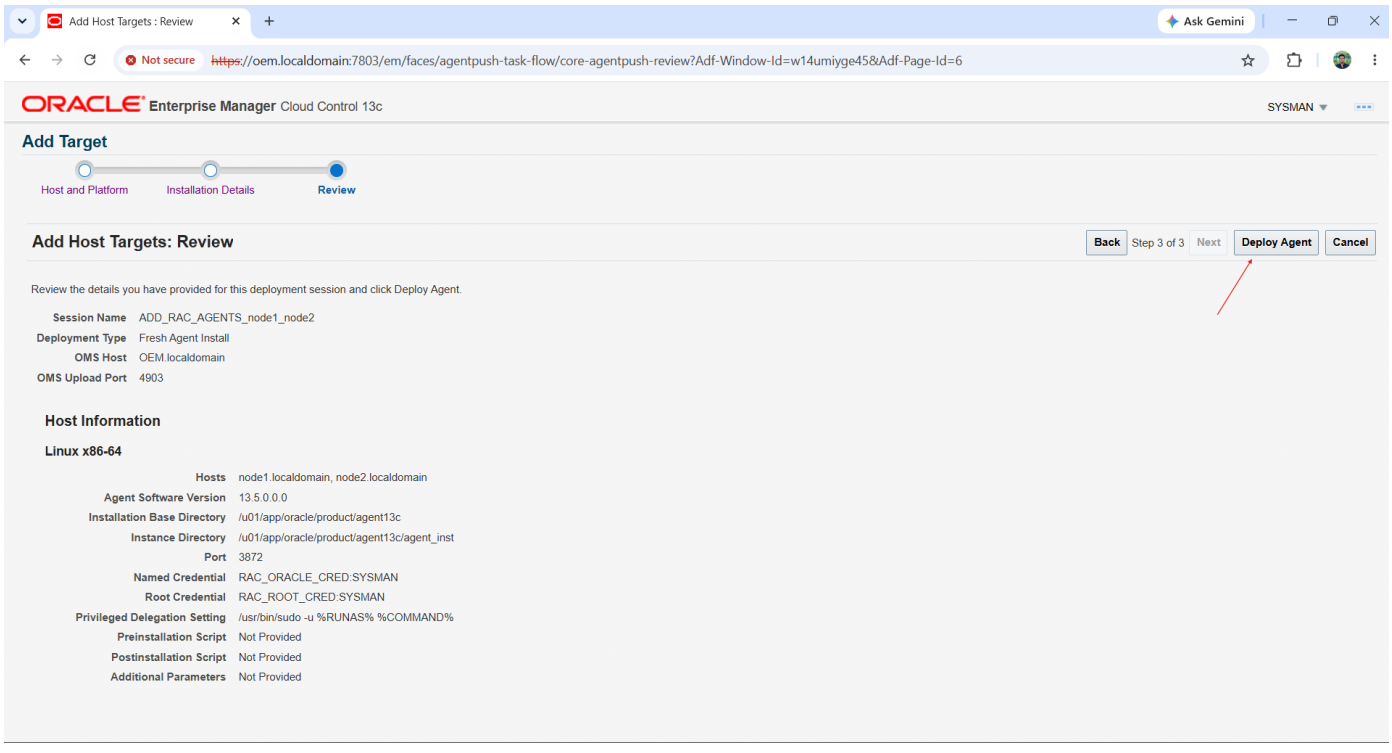
- **Privileged Delegation Setting:** leave default /usr/bin/sudo -u %RUNAS% %COMMAND% (harmless even if unused, since the Root Credential takes priority for root.sh).
- **Port:** 3872

- Click **Next**.



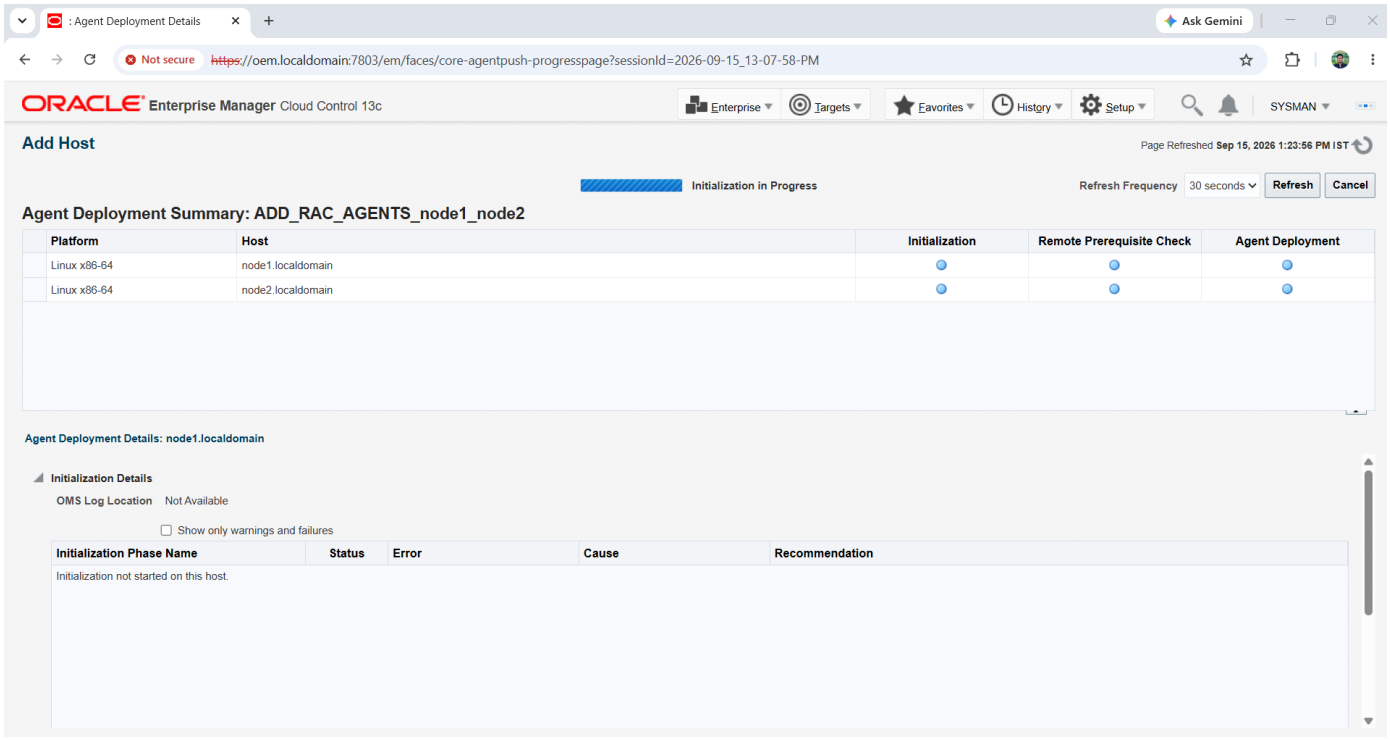
Step 3 of 3 : Review

- Review Session Name, OMS Host, Upload Port, both hosts, install/instance directories, port, and both credentials.
- Click **Deploy Agent**.



Monitoring Deployment

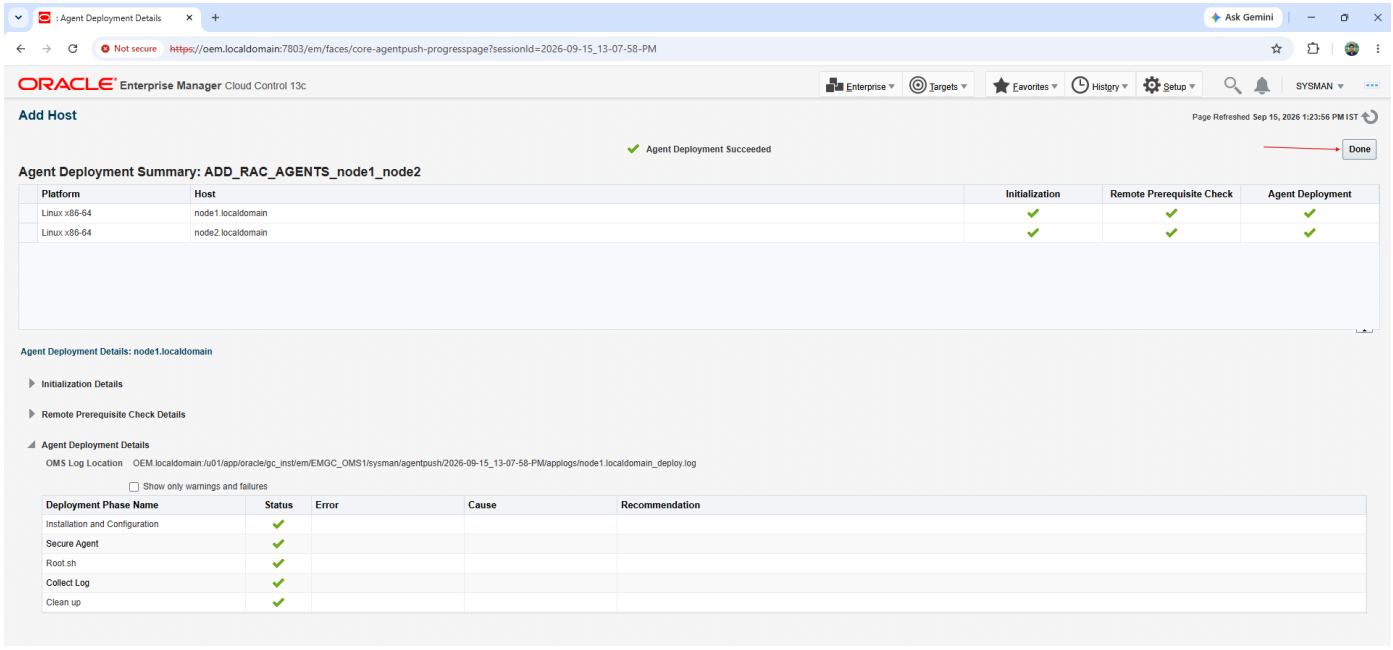
The progress page shows three phases per host : **Initialization → Remote Prerequisite Check → Agent Deployment** — auto-refreshing every 30 seconds.



When complete, it shows:

Agent Deployment Succeeded

with all sub-phases green-checked: Installation and Configuration, Secure Agent, Root.sh, Collect Log, Clean up. Click **Done**.



Verify independently on each node:

Node1:

```
[oracle@OEM ~]$ ssh oracle@node1
Last login: Tue Sep 15 10:50:25 2026 from oem.localdomain
[oracle@node1 ~]$
[oracle@node1 ~]$ /u01/app/oracle/product/agent13c/agent_inst/bin/emctl status agent
Oracle Enterprise Manager Cloud Control 13c Release 5
Copyright (c) 1996, 2021 Oracle Corporation.  All rights reserved.
---------------------------------------------------------------
Agent Version          : 13.5.0.0.0
OMS Version            : 13.5.0.0.0
Protocol Version       : 12.1.0.1.0
Agent Home             : /u01/app/oracle/product/agent13c/agent_inst
Agent Log Directory    : /u01/app/oracle/product/agent13c/agent_inst/sysman/log
Agent Binaries         : /u01/app/oracle/product/agent13c/agent_13.5.0.0.0
Core JAR Location      : /u01/app/oracle/product/agent13c/agent_13.5.0.0.0/jlib
Agent Process ID       : 32281
Parent Process ID      : 32110
Agent URL              : https://node1.localdomain:3872/emd/main/
Local Agent URL in NAT : https://node1.localdomain:3872/emd/main/
Repository URL         : https://OEM.localdomain:4903/empbs/upload
Started at             : 2026-09-15 11:12:48
Started by user        : oracle
Operating System       : Linux version 5.4.17-2136.338.4.2.el7uek.x86_64 (amd64)
Number of Targets      : 3
Last Reload            : (none)
Last successful upload                       : 2026-09-15 11:16:59
Last attempted upload                        : 2026-09-15 11:16:59
Total Megabytes of XML files uploaded so far : 0.37
Number of XML files pending upload           : 0
Size of XML files pending upload(MB)         : 0
Available disk space on upload filesystem    : 86.35%
Collection Status                            : Collections enabled
Heartbeat Status                             : Ok
Last attempted heartbeat to OMS              : 2026-09-15 11:17:04
Last successful heartbeat to OMS             : 2026-09-15 11:17:04
Next scheduled heartbeat to OMS              : 2026-09-15 11:18:04

---------------------------------------------------------------
Agent is Running and Ready
[oracle@node1 ~]$
```

Node2:

```
[oracle@OEM ~]$ ssh oracle@node2
Last login: Tue Sep 15 10:52:02 2026 from 192.168.157.1
[oracle@node2 ~]$
[oracle@node2 ~]$ /u01/app/oracle/product/agent13c/agent_inst/bin/emctl status agent
Oracle Enterprise Manager Cloud Control 13c Release 5
Copyright (c) 1996, 2021 Oracle Corporation.  All rights reserved.
---------------------------------------------------------------
Agent Version          : 13.5.0.0.0
OMS Version            : 13.5.0.0.0
Protocol Version       : 12.1.0.1.0
```

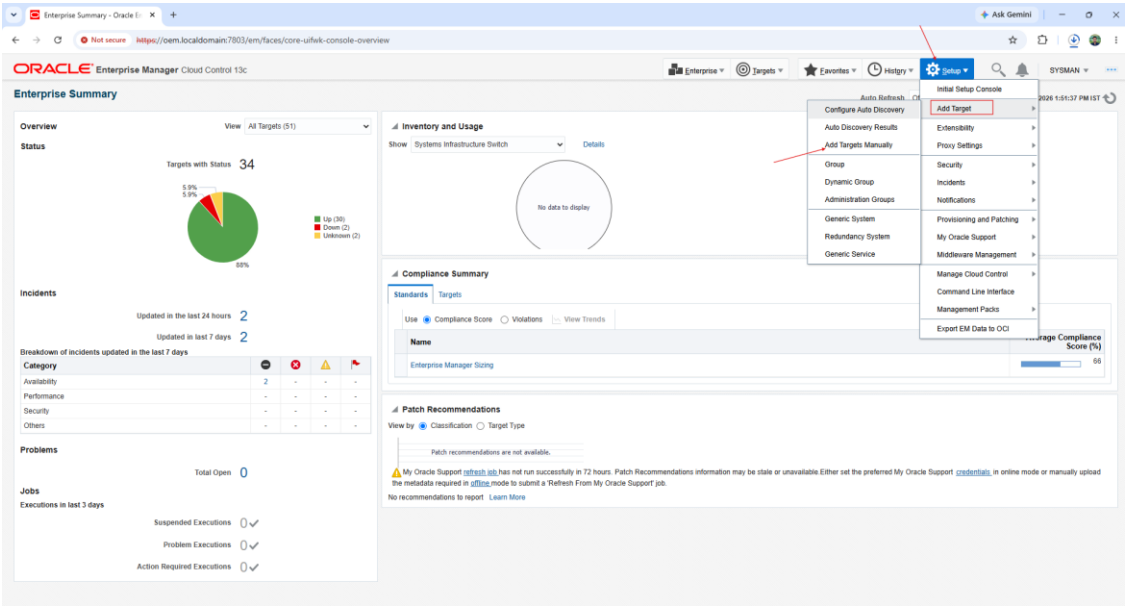
```
Agent Home               : /u01/app/oracle/product/agent13c/agent_inst
Agent Log Directory      : /u01/app/oracle/product/agent13c/agent_inst/sysman/log
Agent Binaries           : /u01/app/oracle/product/agent13c/agent_13.5.0.0.0
Core JAR Location        : /u01/app/oracle/product/agent13c/agent_13.5.0.0.0/jlib
Agent Process ID         : 25108
Parent Process ID        : 24929
Agent URL                : https://node2.localdomain:3872/emd/main/
Local Agent URL in NAT   : https://node2.localdomain:3872/emd/main/
Repository URL           : https://OEM.localdomain:4903/empbs/upload
Started at               : 2026-09-15 11:06:18
Started by user          : oracle
Operating System         : Linux version 5.4.17-2136.338.4.2.el7uek.x86_64 (amd64)
Number of Targets        : 5
Last Reload              : (none)
Last successful upload                       : 2026-09-15 11:17:06
Last attempted upload                        : 2026-09-15 11:17:06
Total Megabytes of XML files uploaded so far : 0.45
Number of XML files pending upload           : 0
Size of XML files pending upload(MB)         : 0
Available disk space on upload filesystem    : 88.57%
Collection Status                            : Collections enabled
Heartbeat Status                             : Ok
Last attempted heartbeat to OMS              : 2026-09-15 11:16:35
Last successful heartbeat to OMS             : 2026-09-15 11:16:35
Next scheduled heartbeat to OMS              : 2026-09-15 11:17:35

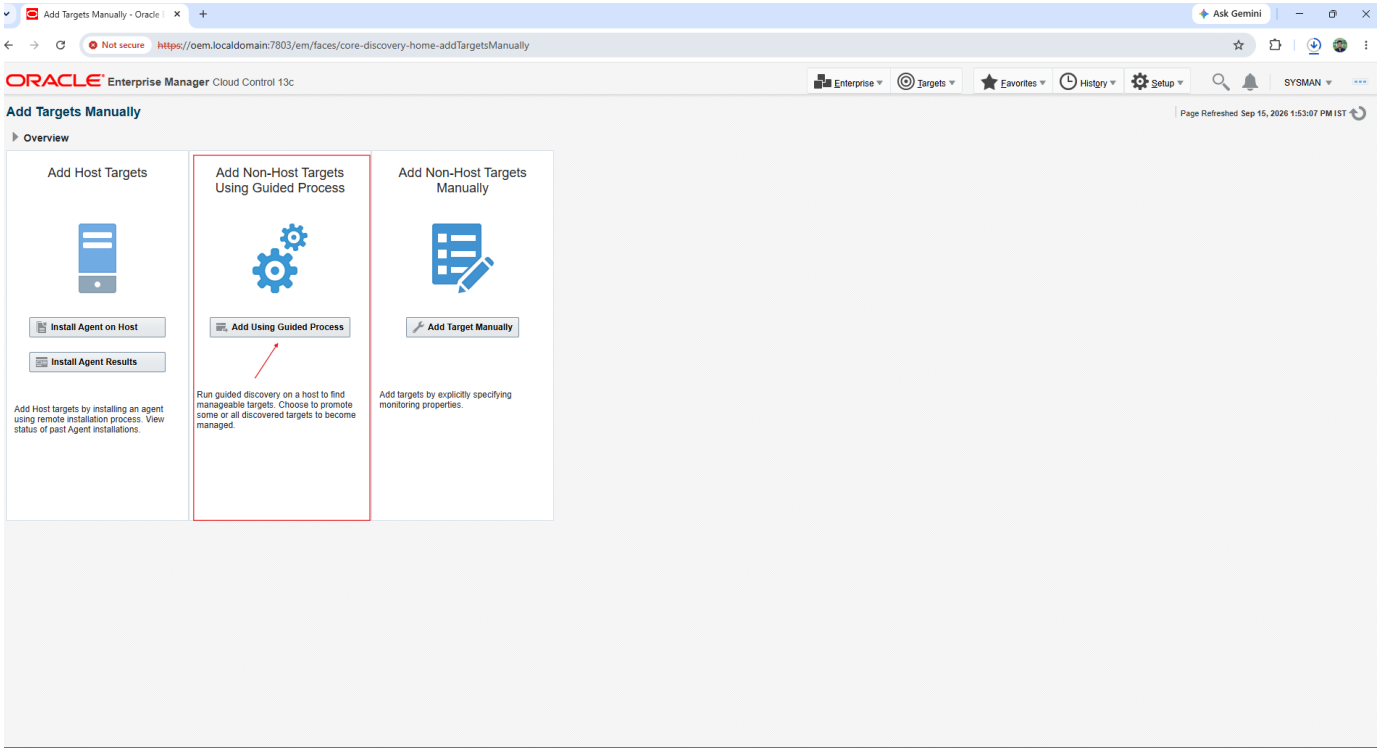
---------------------------------------------------------------
Agent is Running and Ready
[oracle@node2 ~]$
```

PHASE 8 : RAC Cluster Discovery

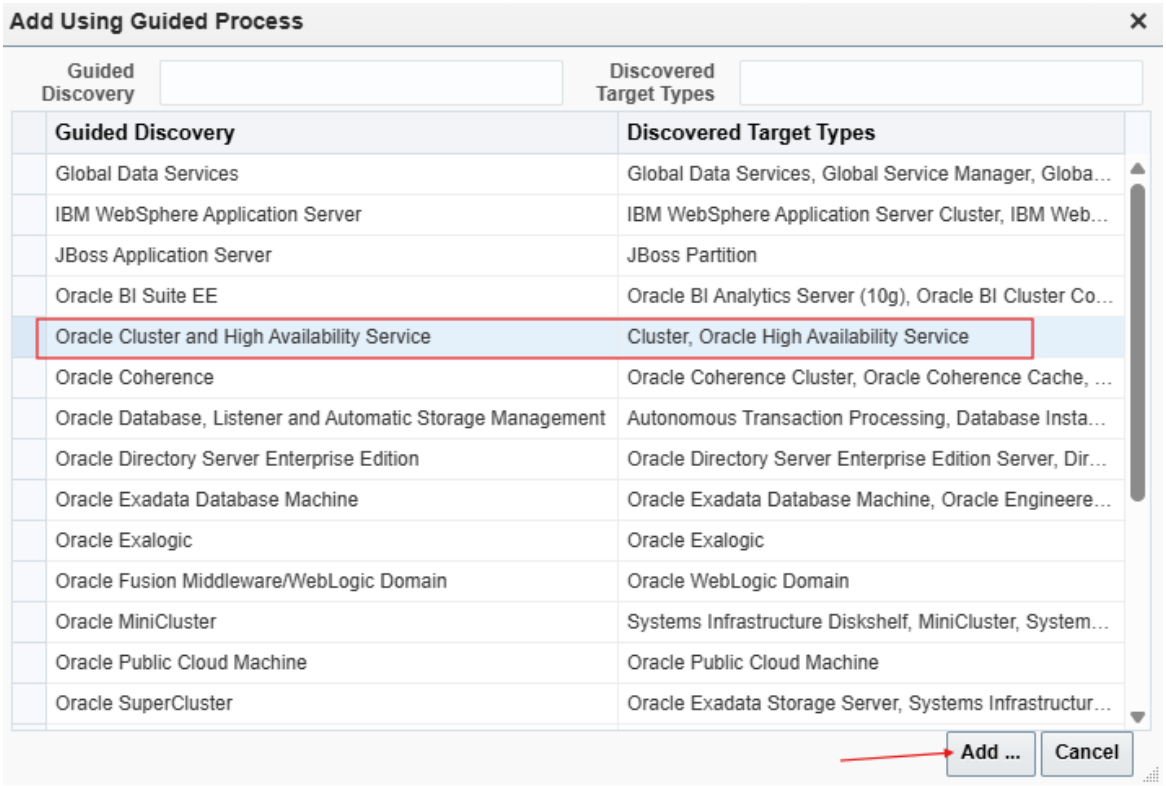
Objective : After the Management Agents are running, the next step is to discover and add the existing Oracle RAC infrastructure to Enterprise Manager.

Navigate: **Setup → Add Target → Add Targets Manually → Add Using Guided Process** (under "Add Non-Host Targets Using Guided Process").



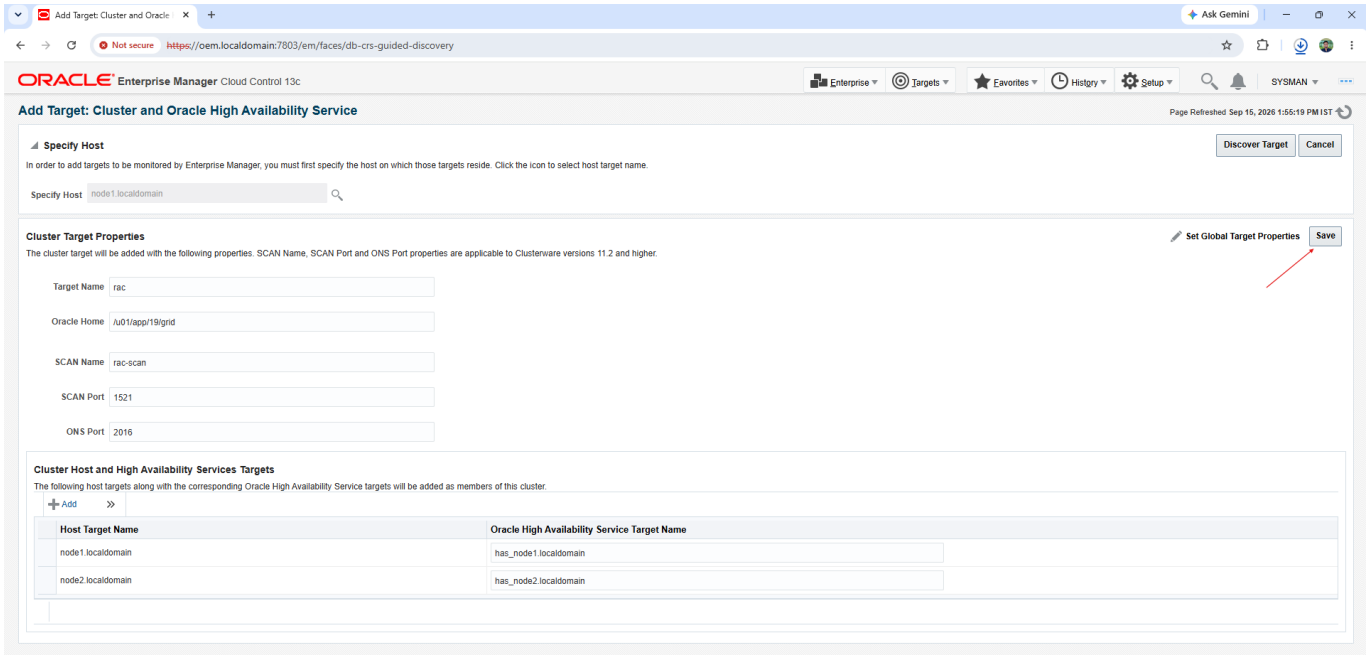


1. In the **Guided Discovery** list, select **Oracle Cluster and High Availability Service** (Discovered Target Types: Cluster, Oracle High Availability Service). Click **Add...**

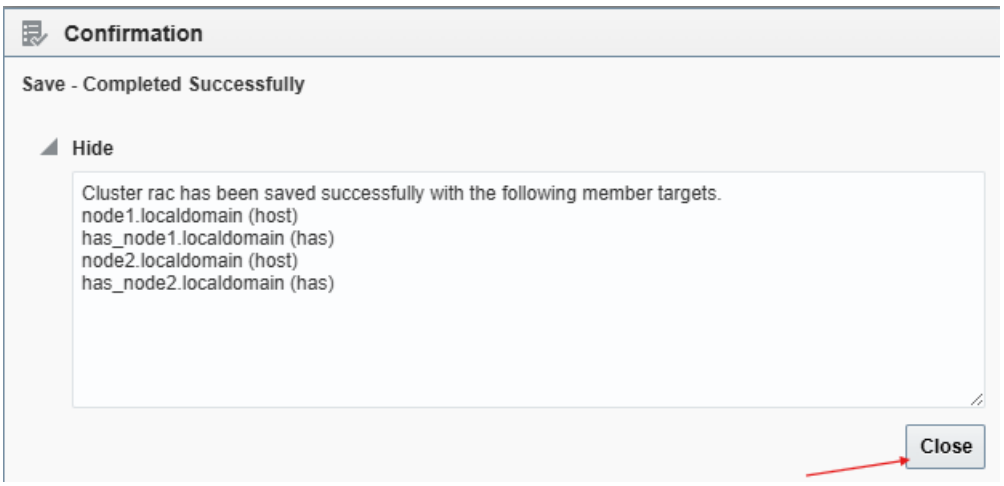


2. **Specify Host** screen: enter/select node1.localdomain in the Specify Host field, then click **Discover Target**.

3. **Cluster Target Properties** : the wizard auto-populates from the OCR:
    - Target Name: rac
    - Oracle Home: /u01/app/19/grid
    - SCAN Name: rac-scan
    - SCAN Port: 1521
    - ONS Port: 2016
    - **Cluster Host and High Availability Services Targets** table auto-fills both nodes: node1.localdomain → has_node1.localdomain, node2.localdomain → has_node2.localdomain
4. Review these auto-detected values, then click **Save**.



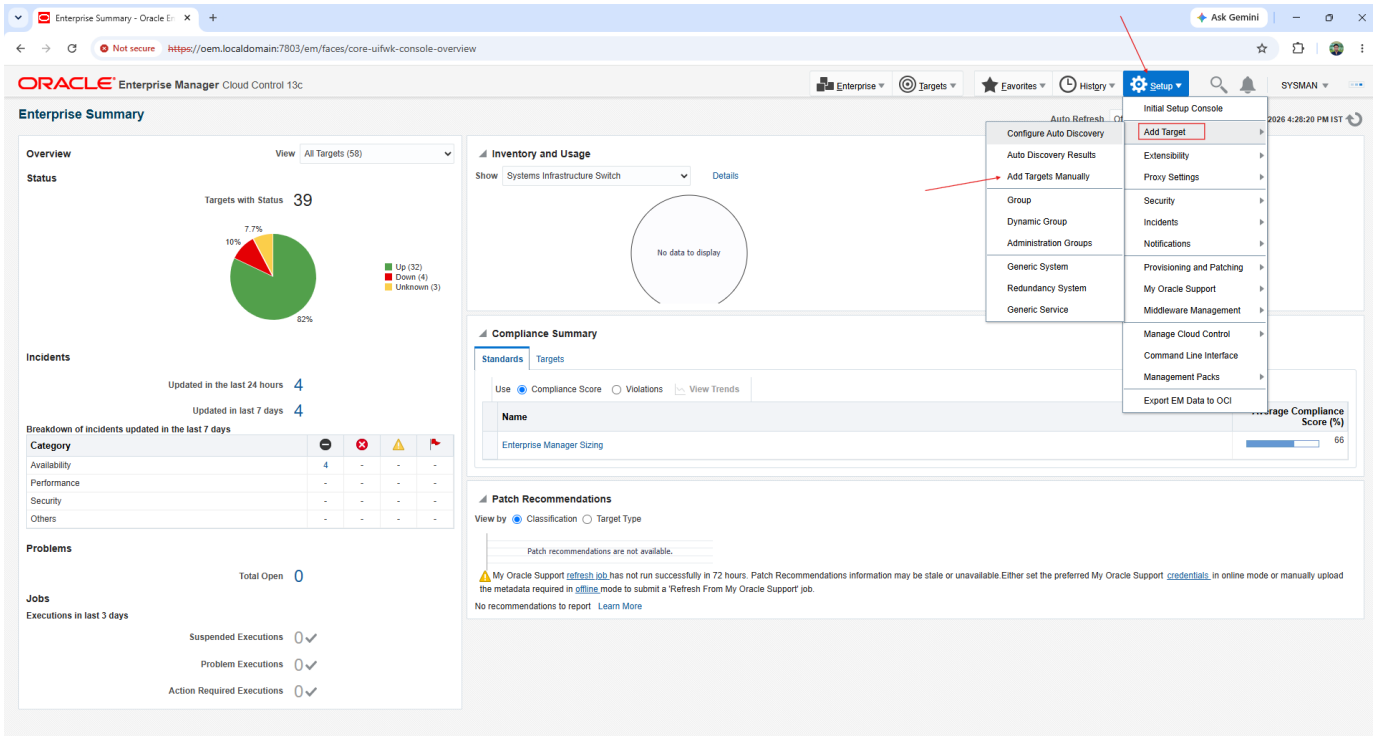
5. A confirmation dialog appears: *"Cluster rac has been saved successfully with the following member targets: node1.localdomain (host), has_node1.localdomain (has), node2.localdomain (host), has_node2.localdomain (has)."* Click **Close**.

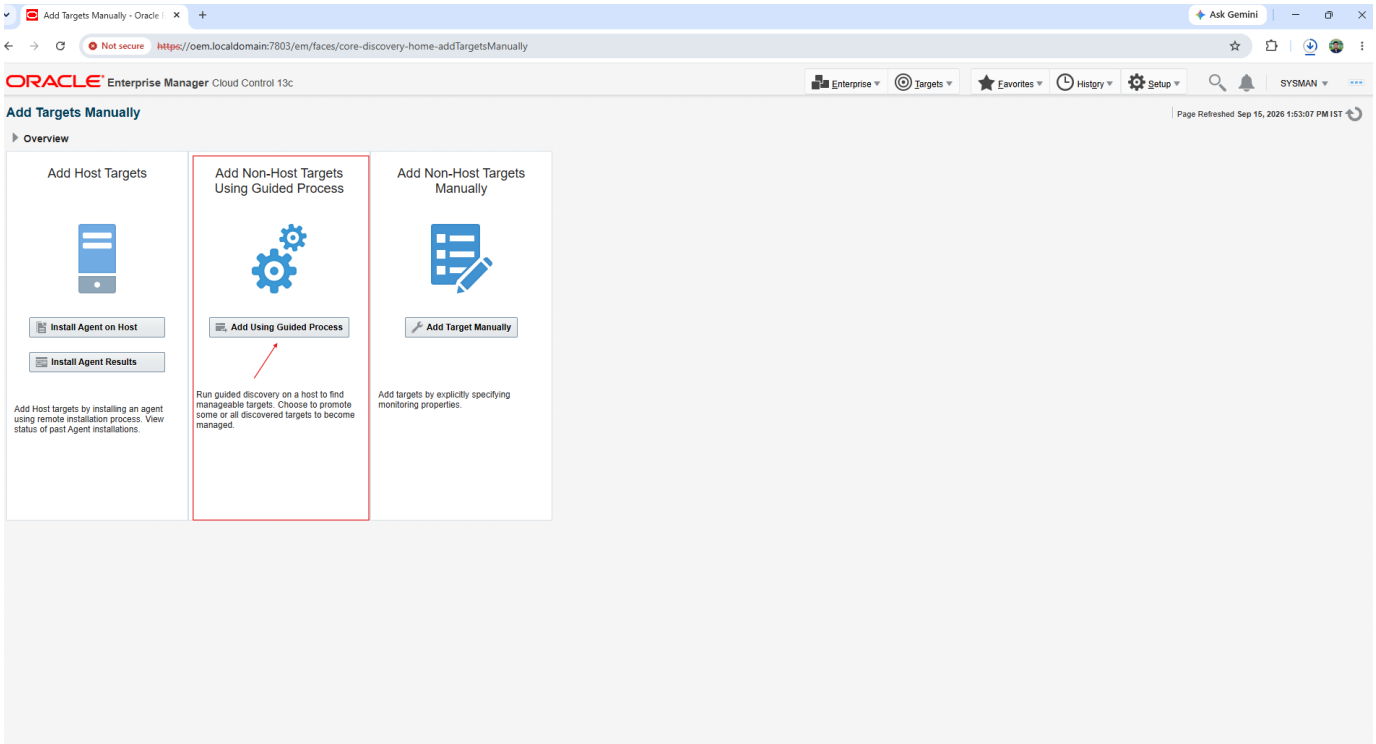


PHASE 9 : Database, ASM, and ListenersDiscovery

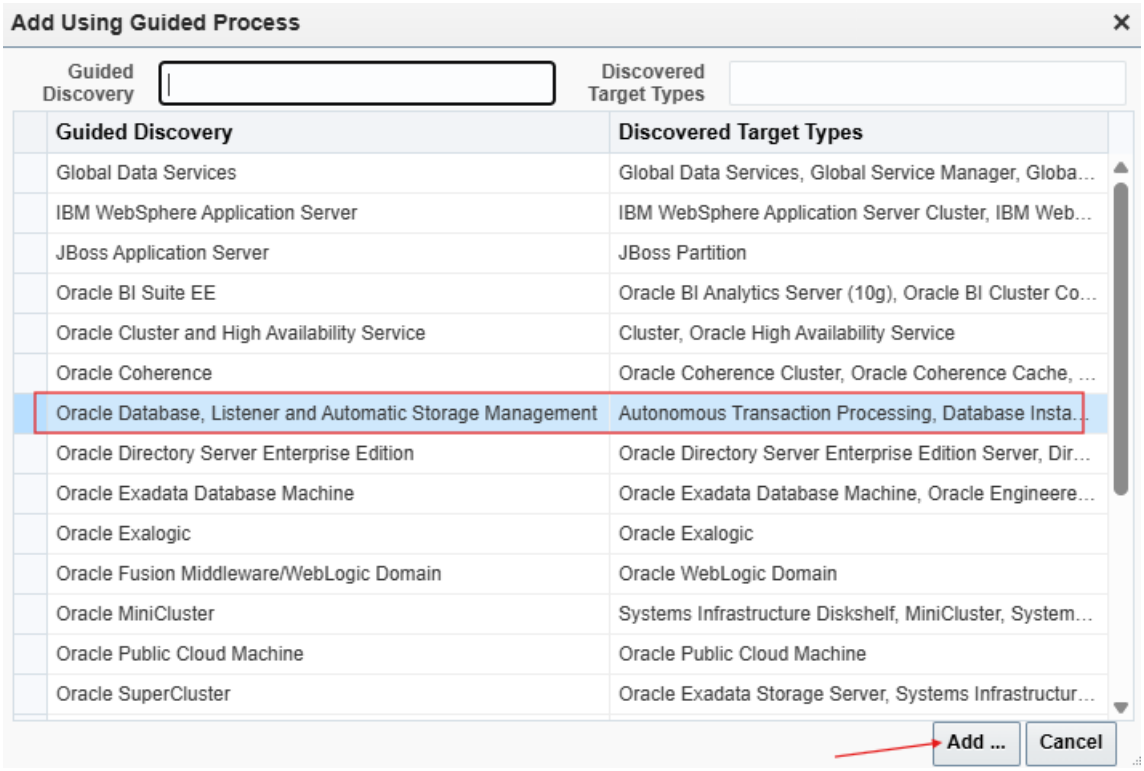
Objective : After the RAC Cluster target has been successfully discovered, the next step is to discover the RAC database, ASM instances, and listener targets associated with the cluster and add them to Enterprise Manager for centralized monitoring.

Navigate to: **Setup → Add Target → Add Targets Manually → Add Non-Host Targets Using Guided Process**.

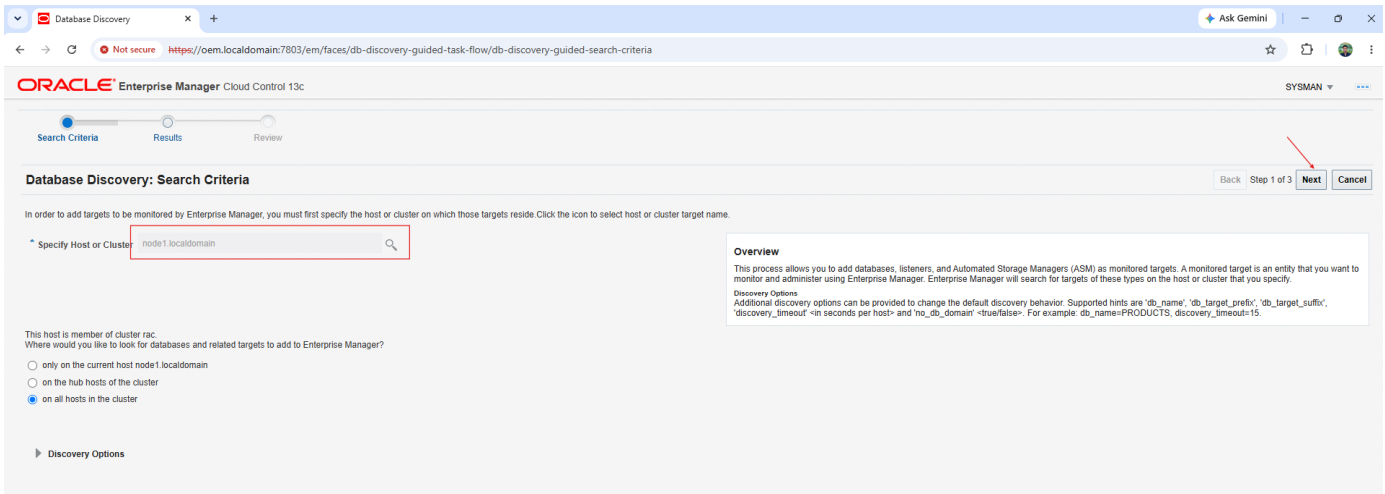




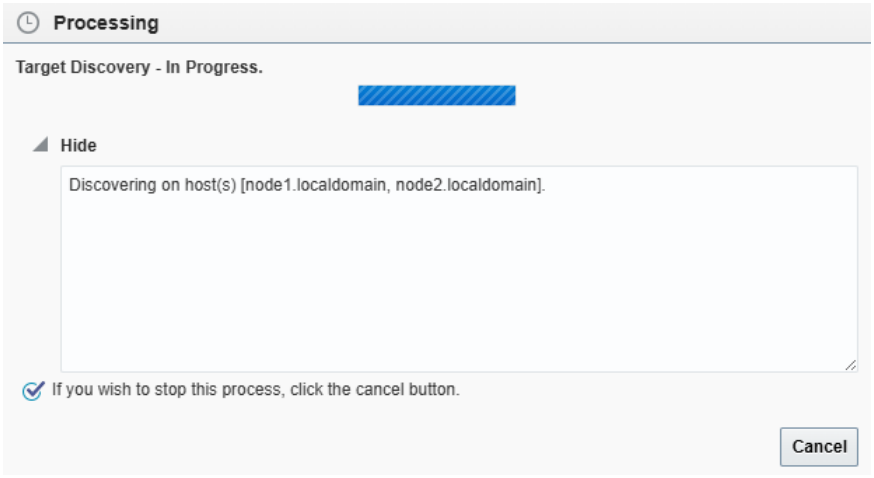
1. In the **Guided Discovery** list, select **Oracle Database** as the target type and click **Add...**.



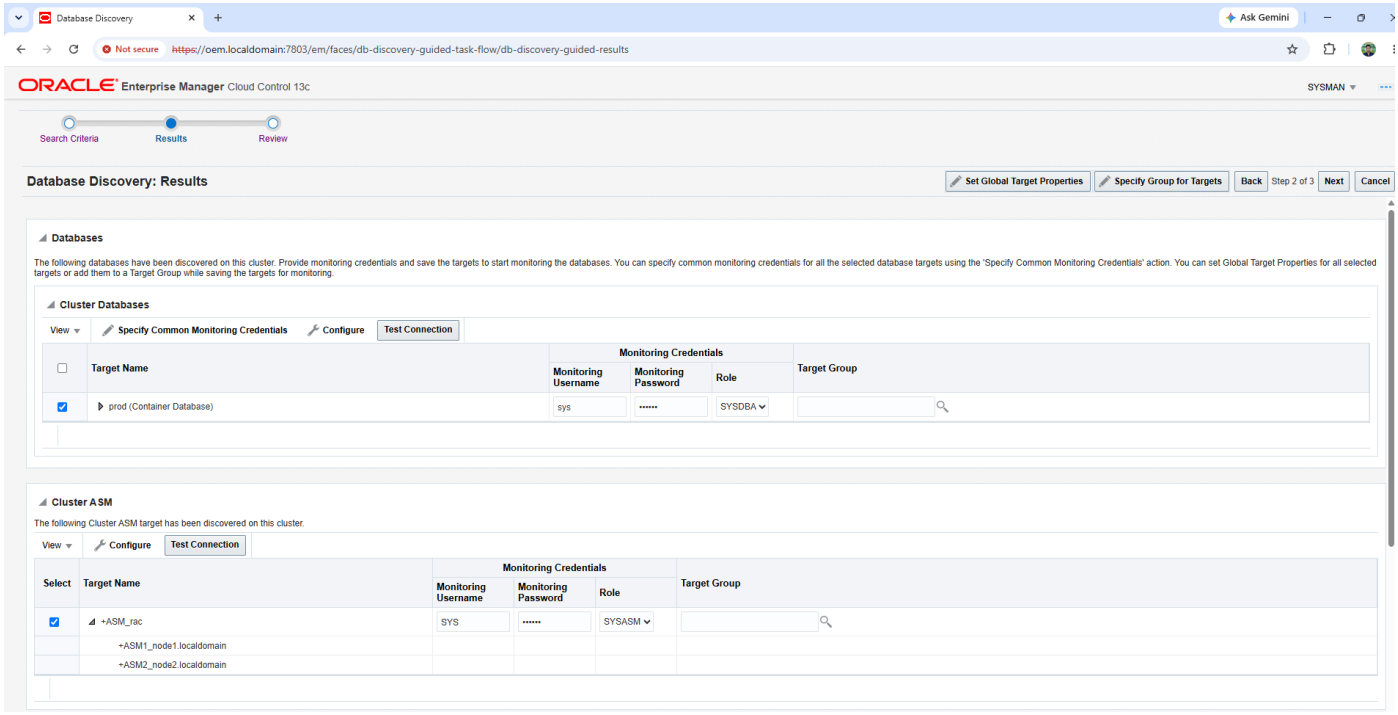
2. In the **Specify Host** screen, enter/select the RAC host where the database and Grid Infrastructure are running, for example **node1.localdomain**, and proceed with **Discover Target**.



3. Enterprise Manager communicates with the selected RAC host and discovers the database-related components associated with the cluster.



4. The **Database Discovery: Results** screen displays the targets discovered from the RAC environment. Review the sections carefully before proceeding.



5. Under **Cluster Databases**, verify that the RAC database is displayed as a single logical database target, for example:

prod (Container Database)

This represents the RAC database as a cluster database rather than treating each RAC instance as an independent database target.

6. Select the discovered database target and provide the required **Monitoring Credentials**.

Enter the configured database monitoring username and monitoring password, then select the appropriate monitoring role, such as **Normal**, according to the account being used.

7. Use **Test Connection** to verify that Enterprise Manager can successfully connect to the database using the supplied monitoring credentials.
8. Under **Cluster ASM**, verify that Enterprise Manager has discovered the ASM instances associated with both RAC nodes.

The discovered ASM targets correspond to the ASM instances running on the RAC nodes, for example:

- **+ASM1_node1.localdomain**
- **+ASM2_node2.localdomain**

9. Select the required ASM targets and provide the **ASM Monitoring Credentials**.

Enter the configured ASM monitoring username and password and select the appropriate role, such as **SYSDBA/SYSASM**, based on the account configured for monitoring.

10. Use **Test Connection** to validate the ASM credentials and confirm that Enterprise Manager can communicate with the ASM instances.
11. Review the **Listeners** section.

Enterprise Manager automatically discovers the listeners associated with the RAC environment, including the node listeners and SCAN listeners.

The discovered targets can include:

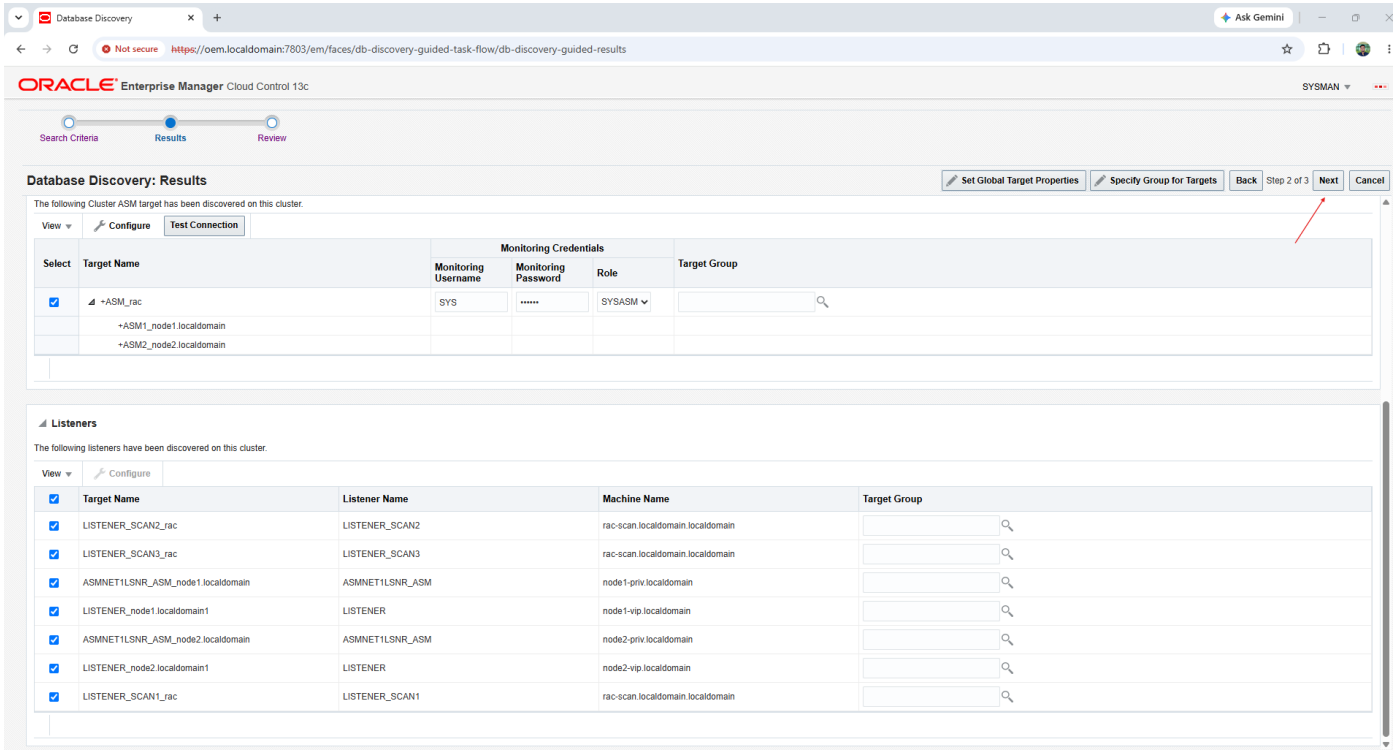
- Node listener on **node1.localdomain**
- Node listener on **node2.localdomain**
- RAC **SCAN listeners**

12. Review the discovered listener names and associated machine names carefully. No database credentials are normally required for the listener discovery itself; Enterprise Manager uses the discovered listener configuration to create the listener targets.
13. Review the complete **Database Discovery: Results** page before continuing.

Confirm that the following components have been discovered:

- RAC Cluster Database **prod**
- ASM instance on **node1**
- ASM instance on **node2**
- Node listeners
- SCAN listeners

14. If all required targets are present and the monitoring credentials have been successfully validated, click **Next**.

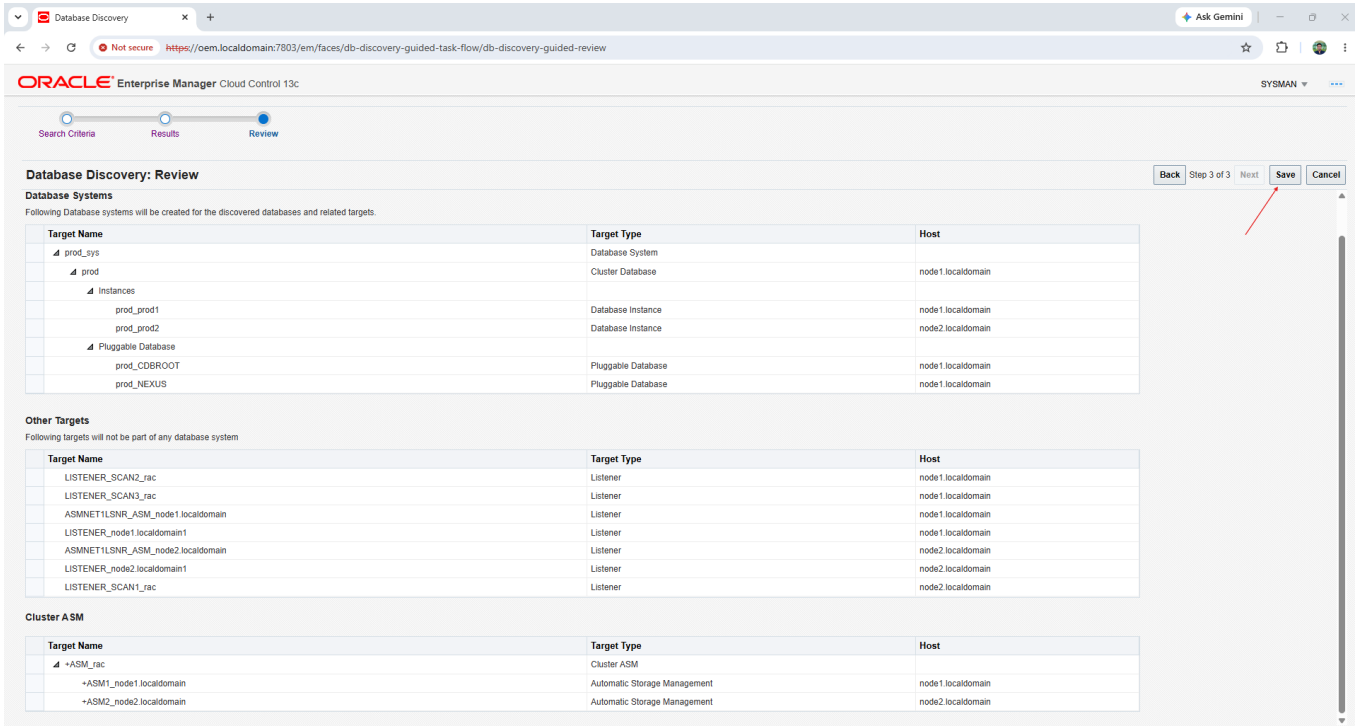


15. The wizard proceeds to the **Review** stage. Review the target information and confirm that the database, ASM, and listener targets belong to the intended RAC environment.

16. Verify that the RAC database is represented as the **cluster database target** and that the individual RAC instances are associated with the appropriate cluster infrastructure.

17. Review the target names, hosts, monitoring credentials, and target relationships one final time. Correct any incorrect information before saving the configuration.

18. Click **Save** to add the discovered database, ASM, and listener targets to Enterprise Manager.

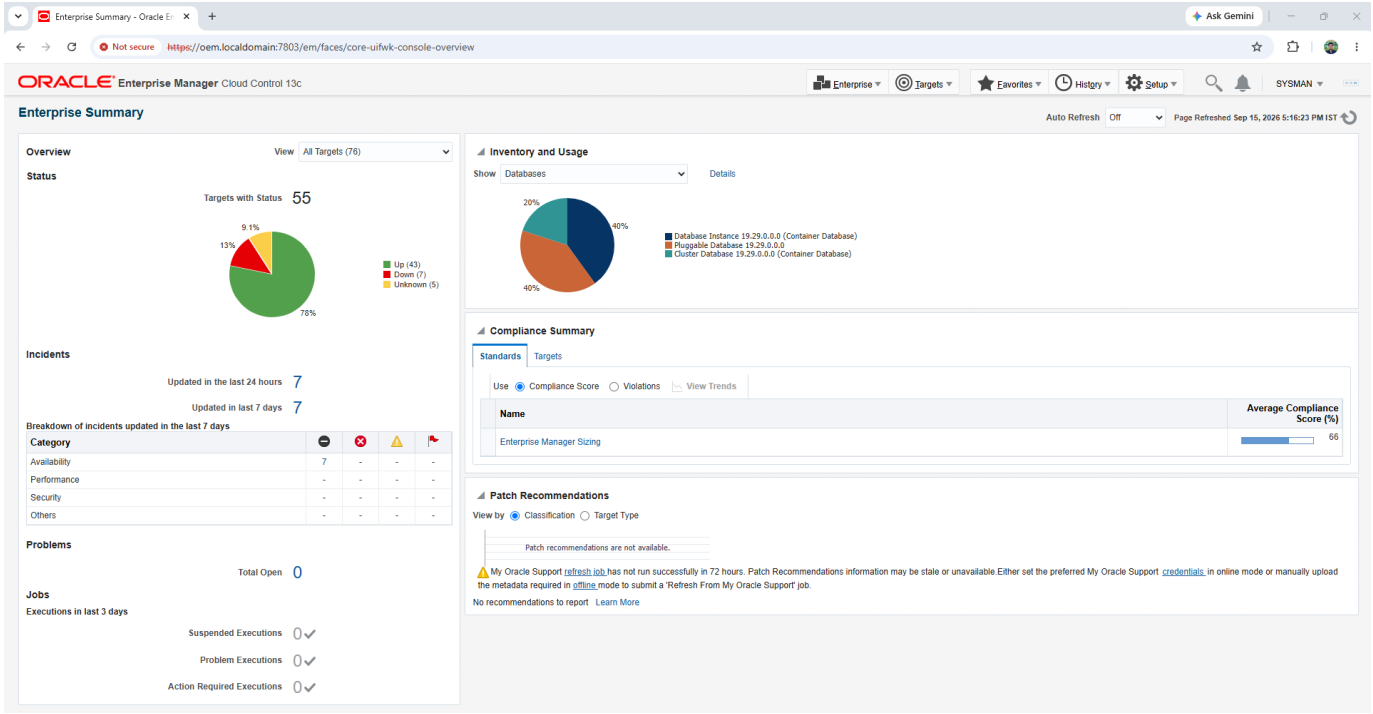


19. After the discovery is completed, navigate to:

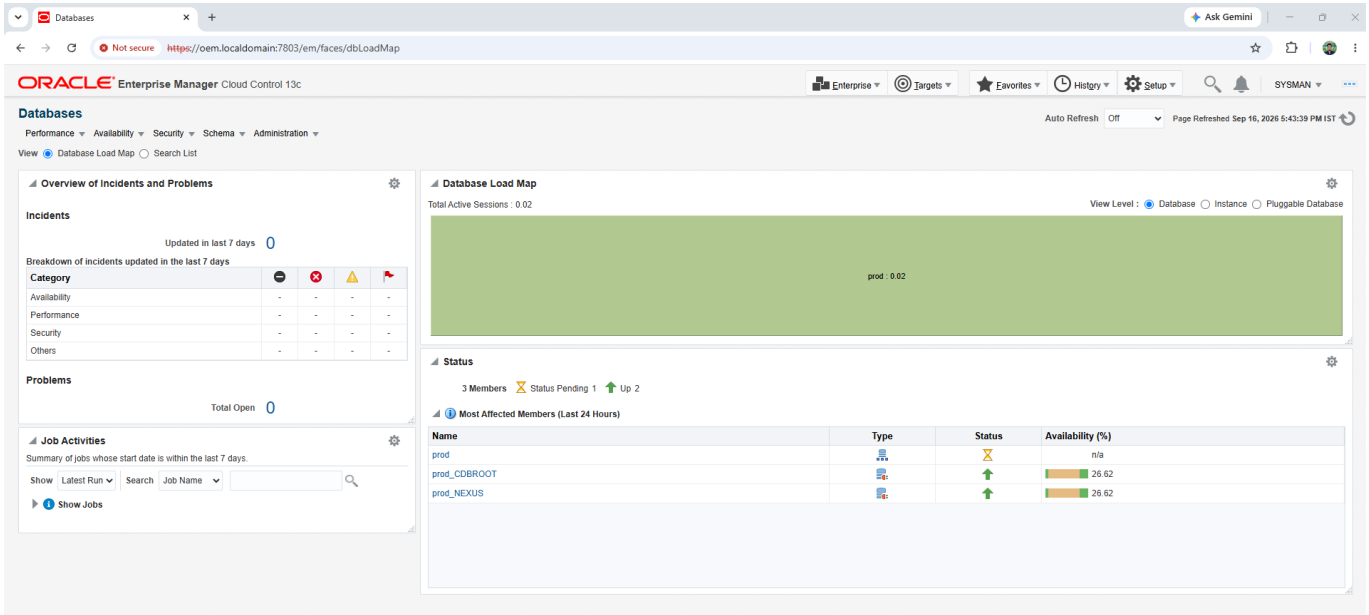
Enterprise Manager → Targets → All Targets

20. Verify that the newly added targets are visible and that their monitoring status is available.

21. Confirm that the RAC hierarchy is represented correctly, with the **RAC Cluster**, **Cluster Database**, **ASM instances**, **listeners**, and RAC nodes available for centralized monitoring.



22. Finally, open the **prod** database target and verify that Enterprise Manager can display the database target information and monitoring data successfully.



Result: The Oracle RAC environment is now fully discovered in Enterprise Manager, including the **RAC cluster, cluster database, ASM instances, node listeners, and SCAN listeners**, providing a centralized monitoring view of the RAC infrastructure.

7. Benefits

• Single console visibility across the 2-node RAC cluster, ASM, listeners, and hosts — no need to log into each node individually to check health

• Historical metric collection enables trend analysis, capacity planning, and faster root-cause analysis during incidents

• Centralized alerting foundation — once configured, EM can proactively notify on tablespace growth, instance down events, listener failures, and more

• Agent-based architecture means monitoring continues to collect and queue data even during temporary connectivity loss to the OMS

• A single Management Agent, once deployed, automatically exposes the host, RAC instance, ASM, and listener as related targets after discovery, reducing repetitive manual setup

• This implementation is now a documented, repeatable procedure — every issue hit has a recorded root cause and fix, cutting rebuild time significantly for future environments

---

8. Conclusion

This implementation successfully established **Oracle Enterprise Manager 13c** for centralized monitoring and management of the Oracle RAC environment. The OEM Management Service (OMS) and Management Repository were configured, the required Management Agents were deployed on both RAC nodes, and the RAC infrastructure was discovered and added to OEM.

The implementation successfully covered:

- Oracle Database 19c software installation for the OEM repository
- Creation and configuration of the **EM Repository Database (emrep)**
- Installation and configuration of **Enterprise Manager 13.5 OMS**
- Configuration and validation of OEM console access
- Management Agent deployment on both RAC nodes
- Discovery of the Oracle RAC cluster and its database
- Discovery and configuration of **ASM instances and disk groups**
- Discovery of RAC listeners and SCAN listeners
- Configuration of monitoring credentials for the database and ASM
- Final validation of OEM targets and monitoring status

With these components in place, the Oracle RAC environment can be monitored from a **single centralized OEM console**, providing visibility into database, ASM, listener, host, and cluster health.

This setup provides a practical foundation for **enterprise database monitoring, performance analysis, availability monitoring, alert management, and proactive administration** of Oracle RAC environments. It also demonstrates an end-to-end approach to deploying OEM 13c and integrating it with an existing Oracle RAC infrastructure.

⚠ Caution / Disclaimer

The information provided in this document is for **educational and learning purposes only**. This guide demonstrates the installation and configuration of **Oracle Enterprise Manager 13c** with an Oracle RAC environment in a controlled lab environment using VMware Workstation. Although the procedures described in this guide were tested successfully during the implementation, always validate the complete procedure in a **UAT or non-production environment** before attempting to implement it on any production system.

This guide is intended to provide **foundational, hands-on understanding of Oracle Enterprise Manager 13c, Management Agents, OMS, the EM Repository, Oracle RAC monitoring, ASM monitoring, and listener discovery**. It should not be considered a substitute for Oracle's official documentation, certified implementation procedures, security standards, or a production deployment checklist.

Before implementing OEM in a production environment, ensure that:

- Proper backups and recovery procedures are available and verified.
- OEM repository sizing and storage requirements are reviewed.
- Required network ports and firewall rules are approved.
- Monitoring credentials and privileges follow organizational security policies.
- Management Agent deployment is tested on representative hosts.
- OMS, repository, and agent compatibility requirements are validated.
- Production changes are properly reviewed, approved, and documented.

Always follow Oracle's official documentation and recommended best practices when deploying Enterprise Manager in production environments. Proceed carefully and validate every configuration step before applying changes to critical systems.

Thank You